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Bachelor's Thesis

**Presence and impact of SNPs and indels in pri-miRNAs
across 1135 *Arabidopsis thaliana* accessions**

In fulfillment of the requirements for the degree of
Bachelor of Science in Bioinformatics

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List of abbreviations

AGO	ARGONAUTE
A. thaliana	Arabidopsis thaliana
BTL	base-to-loop
DCL1	DICER-LIKE 1
dsRNA	double-stranded RNA
ER	endoplasmic reticulum
GFF3	Generic Feature Format 3
HEN1	HUA ENHANCER 1
HYL1	HYPONASTIC LEAVES 1
indel	insertion/deletion
LTB	loop-to-base
MAF	minor allele frequency
MFE	minimum free energy
MIR	MIRNA
miRISC	miRNA-induced silencing complex
miRNA	microRNA
ncRNA	non-coding RNA
nt	nucleotide
pri-miRNA	primary miRNA transcript
RDR6	RNA-DEPENDENT RNA POLYMERASE 6
SBTL	sequential base-to-loop
SE	SERRATE
siRNA	small interfering RNA
SLTB	sequential loop-to-base
SNP	single nucleotide polymorphism
sRNA	small RNA
TAS	trans-acting siRNA locus
tasiRNA	trans-acting siRNA
VCF	variant call format
VIVA	VIsualization of VArants

1 Introduction

Transcriptome analyses revealed that 90% of the eukaryotic genome is transcribed into RNA, but only 1–2% of that corresponds to protein-coding mRNA [4]. The remaining fraction represents non-coding RNAs (ncRNAs), of which two major categories exist in plants based on their regulatory roles [4]. There are housekeeping ncRNAs with a relatively stable level that are essential for fundamental biological processes and there are regulatory ncRNAs, whose expression exhibits spatial and temporal specificity [4]. Three categories of regulatory ncRNAs can be distinguished based on their size: small RNAs (sRNAs) (18–30 nucleotides (nt)), medium-sized ncRNAs (31–200 nt), and long non-coding RNAs (lncRNAs) (>200 nt) [4]. In plants, sRNAs play an important role in the regulation of growth, development, and stress responses [40, 41]. Canonical plant sRNAs include microRNAs (miRNAs) and small interfering RNAs (siRNAs), which are endogenous molecules ranging from 20 to 24 nt in length [4, 41]. The vast majority of plant miRNAs are 21 nt long [41].

1.1 Overview of miRNA biogenesis in plants

This subsection contains an overview of miRNA biogenesis in plants. A schematic presentation can be found in Figure 1.

The biogenesis of miRNAs starts in the nucleus with the transcription of *MIRNA* (*MIR*) genes by RNA polymerase II, producing primary miRNA transcripts (pri-miRNAs) [28, 41]. The greater part of *MIR* genes is located in intergenic and noncoding regions, whereas only a small amount is located in introns of protein-coding genes [41]. Just as with the transcription of protein-coding genes, co-transcriptional 5' capping and 3' polyadenylation are applied on pri-miRNAs throughout the transcription of *MIR* genes, that is regulated by transcription factors [41]. The produced pri-miRNAs are highly variable in length, single-stranded, self-complementary, and can fold into characteristic hairpin structures [28, 41].

Processing of pri-miRNAs takes place within the nucleus and can occur post-transcriptionally or co-transcriptionally [40]. It is mediated by the microprocessor complex and includes a two-step cleavage, first into precursor miRNAs (pre-miRNAs) and then into mature miRNA/miRNA* duplexes, that consist of an miRNA strand and an miRNA* strand [28, 40]. These duplexes have two nucleotide long overhangs at the 3' ends of the two strands [13]. To avoid degradation and assure stability, the miRNA/miRNA* duplexes are 2'-*O*-methylated by HUA ENHANCER 1 (HEN1) at both 3' ends [28, 40]. The core microprocessor complex consists of the RNase III-like endonuclease DICER-LIKE 1 (DCL1), the double-stranded RNA-binding protein HY-POASTIC LEAVES 1 (HYL1), and the zinc-finger protein SERRATE (SE) [28, 40]. HYL1 and SE interact with DCL1 to facilitate precise pri-miRNA processing [13, 41]. Nevertheless, some unprocessed pri-miRNAs may exit the nucleus and be translated into regulatory miRNA-encoded

peptides (miPEPs), because they originate from specific loci or bypass biogenesis due to reduced microprocessor activity [19].

After methylation, the miRNA/miRNA* duplexes are transferred from HYL1 to the ARGONAUTE (AGO) protein AGO1, forming miRNA-induced silencing complexes (miRISCs) [19, 28, 40]. Normally the miRNA* strand is degraded, but the interaction with AGO proteins such as AGO2 or AGO5 is possible to perform regulatory functions in particular organs or under specific stress conditions [40, 41]. MiRISCs subsequently exit the nucleus to the cytoplasm to initiate cell-autonomous RNA silencing [19, 40]. In the course of this, mature miRNAs guide the miRISCs to complementary miRNA target transcripts in the cytosol or on the endoplasmic reticulum (ER) to cleave them, resulting in the repression of gene expression [28, 40]. Moreover, at ER-bound polysomes translation repression can occur by the interaction between AGO1 and the integral membrane protein ALTERED MERISTEM PROGRAM1 (AMP1) [40].

Instead of being transferred to AGO1 in the nucleus, miRNA/miRNA* duplexes can escape the transfer and may be exported to the cytoplasm by HASTY (HST) [19, 40]. In the cytoplasm, these duplexes can either load onto AGO proteins or exit the cell via plasmodesmata to exert non-cell-autonomous activities [19, 40]. Unloaded AGO1 or AGO1 loaded with non-cleavable off-targets are degraded in the cytoplasm [19].

The production of siRNAs can be triggered by particular miRNAs [40]. Thereby, trans-acting siRNA locus (TAS) transcripts are bound by ribosomes, which are held up by SUPPRESSOR OF GENE SILENCING 3 (SGS3) [40]. Furthermore, TAS transcripts are bound and cleaved by miRISCs [40]. One fragment of the cleaved transcript is transformed to double-stranded RNA (dsRNA) by RNA-DEPENDENT RNA POLYMERASE 6 (RDR6) [40, 41]. The subsequent production of 21-nt trans-acting siRNAs (tasiRNAs) is mediated by DCL4 cleavage of the dsRNA [40, 41]. These tasiRNA duplexes are methylated by HEN1 and loaded onto AGO proteins to direct target transcript cleavage [40, 41].

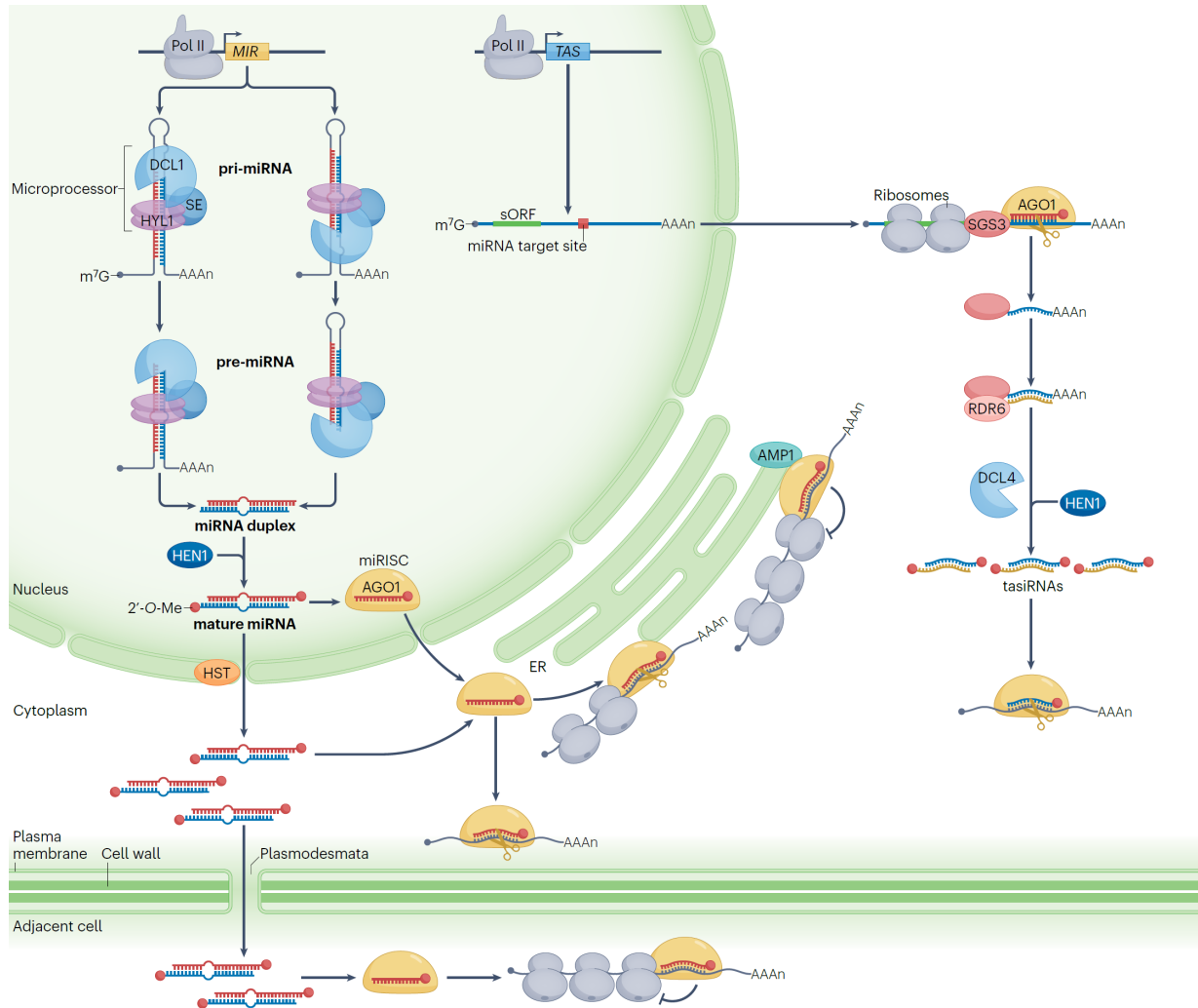


Figure 1: Schematic presentation of miRNA biogenesis and function in plants [40].

MiRNA biogenesis initiates in the nucleus with the transcription of *MIR* genes by RNA polymerase II. The resulting pri-miRNAs can fold into hairpin structures and are processed in a two-step cleavage by the microprocessor complex, either from the base to the loop or vice versa, producing miRNA duplexes. The miRNA duplexes are methylated by HEN1 and subsequently loaded onto AGO1, forming miRISCs, that are transported out of the nucleus to initiate cell-autonomous RNA silencing. Instead of being loaded onto AGO1, miRNA duplexes may be exported to the cytoplasm to either load onto AGO proteins or exit the cell via plasmodesmata to exert non-cell-autonomous activities. Particular miRNAs can initiate the production of tasiRNAs. Thereby, TAS transcripts are cleaved by miRISCs, whereupon one fragment is transformed to dsRNA by RDR6. The following production of tasiRNAs, that can direct target transcript cleavage, is mediated by DCL4.

1.2 Processing mechanisms of pri-miRNAs

In plants, five distinct processing types of pri-miRNAs have been identified (see Figure 2) [28]. In base-to-loop (BTL) processing, an internal loop near the base of the hairpin structure is recognized by DCL1, which leads to the first cleavage located 15–17 nt further towards the upper stem [13, 19, 28]. The second cut occurs approximately 21 nt upstream within the upper stem [28]. In loop-to-base (LTB) processing, DCL1 recognizes an internal loop in the upper stem and cuts 15–17 nt towards the base of the hairpin structure, followed by the second cut located approximately 21 nt downstream [19, 28]. While sequential base-to-loop (SBTL) and sequential loop-to-base (SLTB) processing are also dependent on the 15–17 nt ruler, these sequential mechanisms involve an additional cut 21 nt beyond the second cleavage site [13, 19, 28]. Apart from the 15–17 nt ruler, another smaller internal loop located 9–11 nt away from the first cleavage site affects processing [28]. The 9–11 nt ruler is more commonly observed in LTB processing, in contrast to BTL, where the 15–17 nt ruler is more essential [28]. Bidirectional processing can either start from the base or the loop [28]. Terminal-branched loops are recognized as the central motif in bidirectional processing [28]. Furthermore, bidirectionally processed pri-miRNAs hold a large internal loop prior to the initial cleavage site [28].

For nearly half of all *Arabidopsis thaliana* (*A. thaliana*) pri-miRNAs, the processing type is described. Processing initiates at the base for 36%, at the loop for 35%, and 29% are processed bidirectionally. [28]

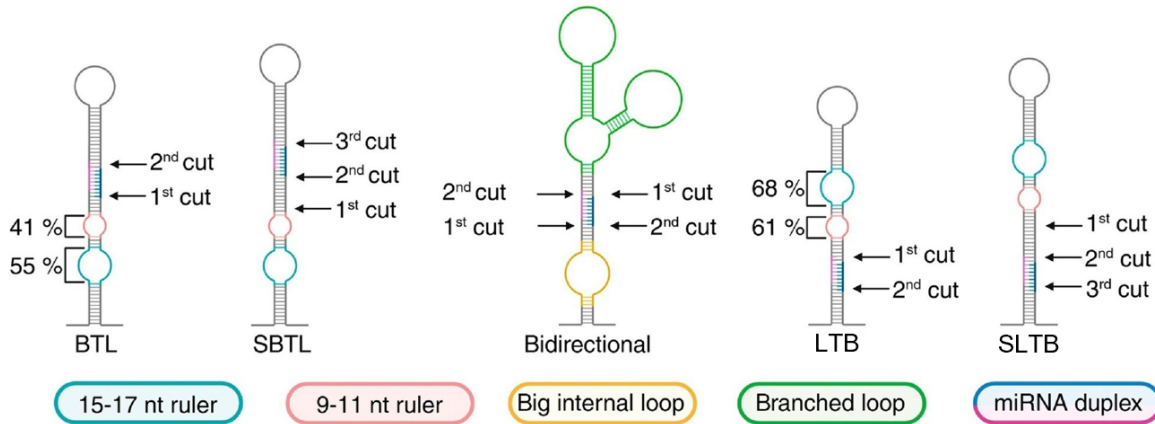


Figure 2: Processing types of pri-miRNAs (modified from [28]).

In plants, five distinct pri-miRNA processing types have been identified: base-to-loop (BTL), sequential base-to-loop (SBTL), bidirectional, loop-to-base (LTB), and sequential loop-to-base (SLTB). The central structural features of the different types are colored. In BTL and SBTL processing, the 15–17 nt and the 9–11 nt ruler are positioned near the base of the hairpin marking the first cleavage site. In LTB and SLTB processing, the rulers also control the first cut but are located in the upper stem. The occurrence of the rulers in the different processing types is presented in percentages. Terminal-branched loops and a large internal loop prior to the initial cleavage site are the key motifs in bidirectional processing. All cuts are approximately 21 nt apart.

Not only the rulers affect processing, but also mismatches in the pri-miRNA [28]. A–C, U–U, and C–U mismatches near cleavage sites have a less severe impact on processing than C–C mismatches [26, 28]. Apart from the specific bases involved, the position of the mismatch in the pri-miRNA can alter processing [28]. In general, a higher frequency of mismatches in the miRNA/miRNA* duplex of pri-miRNAs undergoing a two-step cleavage leads to a decreased interaction energy, resulting in less mature miRNA [26, 28]. Reversely, a lower frequency of mismatches stabilizes the miRNA/miRNA* duplex and increases processing efficiency [26, 28]. In several BTL processed pri-miRNAs, closing mismatches at the base-proximal side of the miRNA/miRNA* duplex decreases second cleavage efficiency without effects on the first cut [28]. In contrast, closing mismatches at the loop-proximal side altered the processing mechanism from BTL to LTB [28].

In addition to the secondary structure, the sequence itself can affect pri-miRNA processing [19, 28]. In the miRNA/miRNA* duplex the presence of GC-rich motifs at position 8–9 and 18–19 is important for precise pri-miRNA processing [19, 28]. Moreover, processing efficiency is increased by a GC content of approximately 52% [28].

Due to the relevance of the secondary structure for pri-miRNA processing, the influence of regulatory factors on pri-miRNA secondary structure is not surprising [13, 28]. For example, chromatin remodeling factor 2 (CHR2) can – despite of its promotion of the transcription of *MIR* genes – directly bind to the pri-miRNA and unwind their secondary structure by its remodeling and helicase activity to repress their processing [13, 17, 28]. In contrast, DExH-box ATP-dependent RNA helicase DExH12 (BRR2A) supports pri-miRNA processing by unwinding the secondary structure to increase the accessibility of cleavage sites to DCL1 and HYL1 [28]. Furthermore, covalent RNA modifications can influence the pri-miRNA secondary structure [19, 28]. The mRNA adenosine methylase (MTA) can bind to pri-miRNAs, initiating m⁶A methylation and subsequently impacting the secondary structure, microprocessor recruitment, and thereby miRNA biogenesis [19, 28]. Inaccurate processing by DCL1, e.g. due to a changed loop shape, can cause the production of alternative miRNAs (also called isomiRs) that have the potential to affect alternative targets [19].

1.3 Variant calling

Small differences between the genomes of individuals of the same species play a crucial role in shaping distinct phenotypes and are therefore of high academic and economic interest [27]. Due to decreasing sequencing costs, whole genome sequencing gained in importance for genetic diversity studies [27, 38]. With the increasing amount of high-quality reference genome sequences, re-sequencing projects grew in significance [27]. In such projects, the main objective is the precise and detailed identification of genetic variations between the reference sequence and a sample [27, 38]. The improvement in sequencing technologies had to go hand in hand with advancements in variant calling algorithms [38].

Variant calling pipelines can roughly be divided into three steps:

1. Read preprocessing
2. Read mapping to a reference sequence
3. Variant calling

Preprocessing of the reads is necessary if the read quality is low, for example due to adapter contamination, overrepresented sequences, or errors during library preparation and sequencing [11, 27]. Such sequencing errors can be identified through quality control and can subsequently be removed with preprocessing [11]. A tool for quality control that visualizes per-base and per-read quality is FastQC, while Cutadapt is used for adapter trimming [11]. Other tools, such as Fastp, can be used for more preprocessing steps including base correction, overlapping analysis, and global trimming [11].

Mapping of the reads to the reference sequence is probably the most crucial step, as the accuracy of all downstream analyses directly depends on the alignment quality [27]. However, the reads are not always mapped to the correct position [27]. Moreover, there is a compromise between mapping speed and the quality of the alignment [27]. Multiple mappers exist that can be categorized based on their indexing methods into hash table-based and Burrows-Wheeler Transformation-based mappers [27, 38]. Not only the mapper itself, but also the quality of the reference sequence determines the performance [27]. Challenges include low-complexity sequences, repetitive regions, collapsed copies of sequences, contaminations, and gaps in the reference genome [27]. A widely used mapper is Bowtie2 [27, 38].

Based on the alignment resulting from the mapping, genomic variants such as single nucleotide polymorphisms (SNPs) or insertions/deletions (indels) can be identified by variant callers [27]. Variant calling precision is enhanced through high sequencing depth to confirm SNPs by multiple reads, a careful study design, and multi-sample joint variant calling to validate genotypes mutually [27]. Common tools such as SAMtools/BCFtools or GATK use a broad range of different methods to detect the variants [27]. The detected variants are subsequently filtered based on quality metrics such as base quality, read depth, and mapping quality to remove uncertain variants [27, 38]. “Hard-filtering” utilizes empirical cutoffs on quality metrics, while machine-learning based filtering methods learn the cutoffs from the specific dataset [38].

The central format to store common sequence variants, such as SNPs, indels, and larger structural variants is called variant call format (VCF) [5, 8]. The VCF file used for the analyses of this thesis was created with the MPI-SHORE [23] and GMI-GATK [7] pipelines by The 1001 Genomes Consortium [32]. To ensure reliability, only high-quality variants called by both pipelines were retained [32]. A detailed description of the VCF file can be found in Section 2.

A key factor in the widespread adoption of the VCF format among bioinformatics software developers is its design as a straightforward and flexible standard that remains easy to read and

interpret [8]. That being said, many tools for comparing, filtering, visualizing, and processing VCF files in additional ways are available [8]. For example, the toolkit `vcflib` includes a wide variety of tools, such as `vcfannotate` to annotate variants that intersect with target regions provided in a BED file [8]. Furthermore, Danecek – who was among the main authors of the VCF format – and his colleagues further developed `BCFtools`, a tool not only used to detect variants, but also to process and analyze VCF files [5, 6]. Moreover, a tool called `VI`sualization of `VA`riants (`VIVA`) was designed to filter, parse and plot VCF files [34]. The usage of `BCFtools` and `VIVA` for this thesis is described in Section 2.

1.4 Objective of this work

This work examines the presence and impact of natural sequence variation in pri-miRNAs across the 1135 *A. thaliana* accessions. To include the highest possible number of pri-miRNAs in the analyses, missing mature miRNAs were predicted. To visualize the presence of sequence variants within the pri-miRNAs, genotype heatmaps were generated. Next, plots per pri-miRNA processing type were created to identify potential patterns between the types. The resulting data were then analyzed statistically to address whether variants are more abundant in pri-miRNA regions compared to the whole genome and to examine whether the observed differences between processing types are statistically significant. After that, haplotype sequences were generated for each accession and pri-miRNA. Their variants were visualized via heatmaps, and the secondary structures of all haplotypes were subsequently predicted. Representative structures were then examined for differences in secondary structure to assess the impact of natural sequence variation.

2 Materials and Methods

This section describes the materials and methods used for the analyses in this thesis. The following subsections are organized to guide the reader through the analytical workflow. Furthermore, all analyses submitted via shell scripts were executed on a high-performance computing (HPC) cluster. Unless otherwise stated, R [25] version 4.5.0 was used. All scripts, raw data, and output files are available via GitHub (<https://github.com/CajusHartmann/Bachelors-Thesis>).

2.1 VCF file from the 1001 Genomes Project

In the course of the 1001 Genomes Project, VCF files were created using the TAIR10 reference genome. For the analyses performed in this thesis, the standard merged group VCF file without invariant positions from <https://1001genomes.org/data/GMI-MPI/releases/v3.1/> was used. The file is called `1001genomes_snp-short-indel_only_ACGTN.vcf.gz` and is 18 GB in size. After the creation, the authors confirmed the overall data quality of the VCF file by validation against one long read and three short read *de novo* genome assemblies. Genotype calling achieved over 98% accuracy at SNP sites, with false positives remaining below 3%. Furthermore, SNPs were misclassified as reference bases in only 1.5% of cases. These estimations do not take into account SNPs overlooked due to nearby structural variants, which short read sequencing struggles to resolve. [32]

```
##fileformat=VCFv4.1
##FILTER=<ID=PASS,Description="All filters passed">
##FILTER=<ID=q30,Description="Quality below 30">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FILTER=<ID=q25,Description="Quality below 25">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT 108
1 21066 . A T 40 PASS DP=23524 GT:GQ:DP 0|0:40:11
2 10505 . T C,G 40 PASS DP=24235 GT:GQ:DP 2|2:34:8
5 40867 . GT G 40 PASS DP=33607 GT:GQ:DP 1|1:40:33
4 40224 . A ATT 40 PASS DP=36644 GT:GQ:DP 1|1:40:23
```

Figure 3: VCF example

The header is from the VCF file used for these analyses. Positions and variants are illustrative. The header contains the version of the VCF file and the meaning of the different abbreviations such as q30, DP or GT in the FILTER, INFO, and FORMAT columns. Moreover, the header shows the names of the columns. The real VCF file contains 1144 columns because of its 1135 accessions, whereas this example only shows accession 108.

The first row after the header indicates a SNP from A to T, however in the presented accession 108 the genotype is homozygous reference as shown by 0|0 in the last column. The second row also depicts a SNP, but with two possible alternate bases. Accession 108 has the genotype G|G as indicated by 2|2. Rows three and four illustrate a deletion and an insertion, respectively.

In Figure 3, the first two columns (**CHROM** and **POS**) represent the position on the genome. Column three (**ID**) does not contain additional information in the used file. Columns four and five (**REF** and **ALT**) hold the reference base(s) and the alternate base(s) as a comma separated list of alternate non-reference alleles. The sixth column (**QUAL**) contains the phred-scaled quality score for the alternate base(s) and the seventh (**FILTER**) indicates the quality filter status. It is always **PASS** because only high-quality variants are retained (see Subsection 1.3). The eighth column (**INFO**) shows the combined read depth across all samples and the ninth (**FORMAT**) contains the format used in the following columns. The last column (**108**) shows the genotype, the quality and the read depth for this specific sample. Allele values within the genotype field are delimited by a slash (/) for unphased data or a vertical bar (|) for phased data. For the reference allele the allele values are 0 and for the first alternate allele 1, for the second 2, and so on. Missing calls are specified as ./.. [5, 33]

2.2 GFF3 file from miRBase

The most important public database for miRNAs is miRBase [14]. Information about miRNAs, such as their sequence, precursors, coordinates on the genome, and additional annotations are available at the miRBase website. Version 22 of miRBase provides 38 589 hairpin precursors (also called pri-miRNAs) from 271 organisms, with 326 hairpins from *A. thaliana*. From these precursors, a total of 428 mature miRNA sequences are annotated in *A. thaliana*. [14]

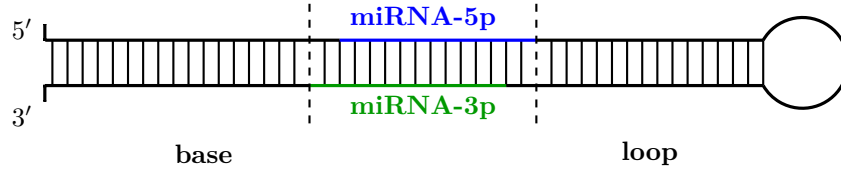
For these analyses, the information about the positions of the mature miRNAs was required. Therefore the **Generic Feature Format 3** (GFF3) file called **ath.gff3** of version 22 was downloaded. It is based on TAIR10 and includes the positions of mature miRNAs and predicted pri-miRNAs on the genome. A total of 95 pri-miRNAs come with mature miRNAs annotated from both the 5' and the 3' arms, whereas the remaining 231 pri-miRNAs have mature miRNAs annotated from only one arm.

2.3 File with pri-miRNA processing types

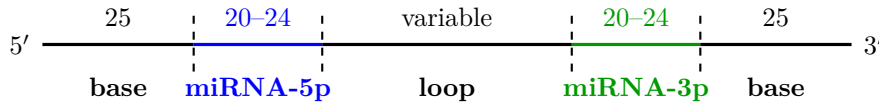
Categorizing pri-miRNAs by processing type was required for the aggregated heatmaps, the variant frequency plots, statistical testing, and evaluating the impact of variants on pri-miRNA secondary structure. Yan et al. identified the initial cleavage sites for 147 *A. thaliana* pri-miRNAs using semi-active DCL1 mutants and degradome-sequencing, additionally providing their processing types [39]. Supplementary Table 2 (Sheet 1) and Supplementary Table 3 from their study were used to group the pri-miRNAs based on processing type [39].

2.4 Organization of pri-miRNA regions on the genome

The structure of pri-miRNAs can be divided into a double-stranded stem and a terminal loop [28]. The double-stranded stem contains the flanking base regions as well as the miRNA-5p and miRNA-3p sequences, while the loop connects the two strands of the RNA structure [28]. For the analyses performed in this thesis, the flanking base regions were set to 25 nt each to include the most important regions involved in pri-miRNA processing (see Subsection 1.2). The genomic organization of these regions and pri-miRNA structure are shown in Figure 4.



(a) This figure illustrates the simplified structure of pri-miRNAs, showing the double-stranded stem – containing the base and miRNA regions – and the terminal loop. The miRNA-5p is close to the 5' end, whereas the miRNA-3p is close to the 3' end of the RNA strand.



(b) The organization of the pri-miRNA regions on the coding DNA strand is shown. The numbers indicate the length of the regions in nucleotides. The length of the loop varies between pri-miRNAs.

Figure 4: Structure and genomic organization of pri-miRNAs. Not drawn to scale.

2.5 Predicting missing mature miRNAs

As described in Subsection 2.2, only 95 pri-miRNAs from the miRBase GFF3 file come with mature miRNAs annotated from both arms. For the remaining 231 pri-miRNAs, the missing mature miRNAs were predicted with StarSeeker (version 1.0) [20] in the course of this work. StarSeeker requires input files containing precursor and mature sequences to predict mature miRNAs considering the secondary structure of the pri-miRNAs and the characteristic two nucleotide overhangs generated by DCL1 [20]. These files (`hairpin.fa` and `mature.fa`) were retrieved from miRBase version 22 [14]. They were subsequently filtered with R [25] to only retain the sequences from *A. thaliana* and to exclude the 95 pri-miRNAs already annotated from both arms with their mature miRNAs.

After running StarSeeker, miR829-3p.2, miR829-5p, miR1886.2, and miR1886.3 were removed from the output because they were missed by the initial filter although they are not predicted but already present in miRBase as annotations for both arms. Furthermore, *MIR156i* was removed from the output as no prediction could be made due to the unusual structure of the pri-miRNA.

Additionally, for *MIR447b*, *MIR858a*, and *MIR5020a* two predictions were generated, one of which was not located across from the already annotated mature miRNA. Consequently, the non-opposite predictions were discarded.

The output of StarSeeker contains the sequences of the predicted mature miRNAs. For further analyses the genomic coordinates of these mature miRNAs were calculated. Following, the 95 pri-miRNAs that come with mature miRNAs annotated from both arms are called non-predicted miRNAs and the remaining 231 pri-miRNAs are called predicted miRNAs.

2.6 Filtering the VCF file

To simplify the following analyses regarding storage consumption and processing time, the 18 GB VCF file from the 1001 Genomes Project described in Subsection 2.1 was filtered. BCFtools (version 1.23.1) [6] was used to discard all variants outside of miRNA regions. These regions are defined as described in Figure 4b and the files containing these regions were generated using R [25] with the package dplyr (version 1.1.4) [37]. For the non-predicted miRNAs the bases were set to 50 nt each for filtering because initially this value was chosen for the length of the bases.

First, an index of the VCF file was created. Then the file was filtered so that only variants within the non-predicted miRNAs are retained. After that, the original file was filtered again to receive another file with the variants within the predicted miRNAs. The commands for the non-predicted miRNAs were as follows:

```
bcftools index 1001genomes.vcf.gz
bcftools view 1001genomes.vcf.gz -O v -o miRNA_positions.vcf -R regions.tsv
```

The option `-O v` sets the output format to an uncompressed VCF file, `-o` specifies the output file, and `-R` defines the regions in which variants are retained. For the predicted miRNAs the output file was named `miRNA_positions_pred.vcf` and the `regions_pred.tsv` file that contains the regions of the predicted miRNAs was passed to `-R`.

2.7 Genotype heatmaps

VIVA is a tool that combines the functionality of multiple tools into a single command to filter, parse and plot VCF files [34]. To specify the positions that VIVA should consider, it requires a comma separated list of genomic coordinates as input [34]. VIVA (version 0.4.0) [34] was executed using Julia (version 1.4.1) [1] separately for every pri-miRNA via array jobs, using their specific genomic coordinates. As with the filtering of the VCF file, the bases of the non-predicted miRNAs were set to 50 nt each. The command used to start VIVA for the non-predicted miRNAs is shown below:

```
viva -f "miRNA_positions.vcf" -l "$NAME.csv" -s pdf -o "$NAME" \
--heatmap genotype,read_depth -n --avg_dp sample,variant
```

The option `-f` specifies the input VCF file, `-l` the positions to consider, `-s` the plot format, `-o` the output directory, `--heatmap` the plot types to create, `-n` the saving of numeric arrays, and `--avg_dp` the visualization of average read depths. `"$NAME"` represents the name of the current pri-miRNA. For the predicted miRNAs, only the input VCF file was exchanged. It should be noted that in the `define_geno_dict()` function of the `vcf_utils_complete.jl` script from VIVA, the array called `homo_variant` was expanded with "7/7" and "8/8" to allow VIVA to handle eight different variants. Without this adjustment, one of the array jobs failed.

A limitation of the genotype heatmaps generated by VIVA is that they only visualize variant sites. For instance, a 150 nt long pri-miRNA containing three variants at different positions is represented by a plot displaying only those three positions for the 1135 accessions. Therefore, genotype heatmaps of the entire miRNA regions were generated in the course of this thesis using R [25] with the packages `plotly` (version 4.12.0) [31] and `webshot2` (version 0.1.2) [3]. These heatmaps are based on the numeric arrays of categorical genotype values created by VIVA. Instead of storing specific nucleotide identities, the numeric arrays encode the genotypes only as numerical values, where 0 represents missing data, 1 homozygous reference, 2 heterozygous variant, and 3 homozygous variant. Positions of the bases, the mature miRNAs, and the loops were calculated to mark them in the heatmaps. All bases are 25 nt long, both for the non-predicted miRNAs and for the predicted miRNAs. Additionally, filtered heatmaps were generated that only visualize variants with a minor allele frequency (MAF) greater than or equal to 0.005 and a missing rate less than or equal to 0.5.

2.8 Plots per processing type

With the files from Yan et al. [39], pri-miRNAs were grouped based on processing type (see Subsection 2.3 and Table 2). All analyses described in this section were performed on the non-filtered data and separately for the non-predicted and the predicted miRNAs.

2.8.1 Aggregated heatmaps

To get an overview of the heatmaps and to search for patterns across the processing types, aggregated heatmaps were created per processing type using R [25] with the packages `plotly` (version 4.12.0) [31], `webshot2` (version 0.1.2) [3], and `tidyverse` (version 2.0.0) [36].

Every heatmap is internally represented as a matrix. These matrices always have 1135 rows for the accessions, but the number of columns varies depending on the length of the pri-miRNA. For the generation of aggregated heatmaps, the number of columns has to be identical and was

therefore scaled. As seen in Subsection 2.4, the bases are always 25 nt long. For this reason they did not have to be scaled. The mature miRNA regions and the loops were scaled separately because they vary in length. To exclude the regions where the rulers described in Subsection 1.2 occur, 20 nt at the start and end of the loops were not scaled. Loops were scaled to the median length of the group (see Table 1) minus 40 and mature miRNA regions were scaled to 21. To increase the length of regions, nearest-neighbor resampling was applied to keep the value range. To decrease the length, block aggregation using the maximum operator was used to avoid a loss of SNPs.

Table 1: Median length of the loops per processing type for the non-predicted and predicted miRNAs

Processing type	Non-predicted	Predicted
Bidirectional	61	43
BTL	51	51
LTB	42	58
SBTL	62	100
SLTB	114	132

To receive values between 0 and 1 in the aggregated heatmaps at each position, the values of the heatmaps were modified before matrix addition. 0 (missing data) was set to NA (indicates missing values in R), 1 (homozygous reference) to 0, and 2 and 3 (heterozygous and homozygous variants) to 1. To create aggregated heatmaps, the matrices within a group were added element-wise and subsequently divided by the group size. Consequently, a position in the aggregated heatmap reaches the maximum value of 1 only when every heatmap in that group carries a variant at that position. The minimum value of 0 indicates that no heatmap carries a variant at that position. Values between 0 and 1 represent the proportion of heatmaps carrying a variant (e.g., 0.5 for half of the heatmaps). It should be noted that in R [25], addition involving missing values (NA) is NA. However, for the generation of the aggregated heatmaps addition between NA and a numeric value was treated in such a way that it returns the numeric value. Consequently, NA values only persist in the aggregated heatmaps if all values at a specific position are missing, in which case they are displayed transparently.

2.8.2 Variant frequency plots

Apart from the aggregated heatmaps, variant frequency plots were created per processing type with R [25] and the package tidyverse (version 2.0.0) [36] based on the genotype matrices created by VIVA. They display the variant frequency across the 1135 accessions on the y-axis and the position in the pri-miRNA on the x-axis. The regions are scaled in the same manner as for the aggregated heatmaps. A frequency of one indicates that the variant at this specific position

occurs in all 1135 accessions. The darkness of the lines indicates the number of miRNAs in which the variant occurs.

2.9 Statistical analyses

All analyses described in this section were performed separately for the non-predicted and the predicted miRNAs.

2.9.1 Permutation test

To examine if variants are more abundant in pri-miRNA regions compared to the whole genome, the R (version 4.5.1) [25] packages `regioner` (version 1.42.0) [9] and `GenomicRanges` (version 1.62.1) [16] were used to investigate if miRNA regions overlap variants more often than expected by chance. To generate a file with the genomic coordinates of all SNPs and indels from the whole genome, `BCFtools` (version 1.23.1) [6] was applied as follows:

```
bcftools query -f '%CHROM %POS\n' 1001genomes.vcf.gz -o chrom_pos_all.vcf
```

The option `-f` specifies the columns to be extracted, `1001genomes.vcf.gz` is the 18 GB VCF file for the whole genome, and `-o` determines the output file. This output file was subsequently used to perform the permutation test with 1000 permutations using `regioner` [9]. The result of the permutation test is a comma-separated file containing the names, p -values, z -scores, and alternative hypotheses of the miRNAs. The p -values indicate the probabilities of the overlaps assuming the null hypothesis that the overlap occurs by chance. The z -scores quantify the strength and direction of the associations by measuring how many standard deviations the observed overlaps deviates from the expected mean, and the alternative hypotheses specify the expected direction of the effect. Finally, miRNAs with p -values less than 0.05 were filtered.

2.9.2 Kruskal-Wallis test

Additionally, differences between the processing types regarding the number of SNPs and indels in the regions of the pri-miRNAs were examined statistically. Therefore, input files containing the names of the miRNAs, the specific regions with their lengths, the number of variants, and the processing types were compiled. First, a Shapiro-Wilk test [29] was performed in R [25] to assess whether the data were normally distributed. Since the data were not normally distributed and unpaired, a Kruskal-Wallis test [15] was subsequently applied for each region. Furthermore, the statistical analyses were repeated on a filtered dataset in which rows with variant counts greater than or equal to 700 were excluded.

2.10 Haplotype sequences

The genotype heatmaps described in Subsection 2.7 highlight the variant positions for all 1135 accessions per miRNA not considering the nucleotide identities. This means, that many variants in a lot of accessions could be visualized, although the accessions all have the same sequence. To address this problem, the actual DNA sequence for every accession and pri-miRNA was extracted, and all haplotypes were visualized via heatmaps.

First, the filtered VCF files (see Subsection 2.6) were filtered again with a MAF of 1% to only include variants in the haplotype analyses that occur in at least 1% of the accessions. To implement this, BCFtools (version 1.23.1) [6] was used as follows for the non-predicted miRNAs:

```
bcftools view -i 'MAF >= 0.01' miRNA_positions.vcf -O v \
-o miRNA_positions_filtered_MAF0.01.vcf
```

The option `-i 'MAF >= 0.01'` restricts the output to variants with a MAF greater than or equal to 1%. This is followed by the input file specification, while `-O v` sets the output format to an uncompressed VCF file, and `-o` defines the output file. For the predicted miRNAs, `miRNA_positions_pred.vcf` was used as the input file, and the output was saved as `miRNA_positions_pred_filtered_MAF0.01.vcf`.

After the filtering, `samtools faidx` was applied on every miRNA region using the TAIR10 reference genome to obtain the DNA reference sequence that is transcribed into the pri-miRNA. Subsequently, `bcftools consensus` was used to generate one FASTA file per miRNA with the sequences of all accessions. When the miRNA is located on the reverse DNA strand, SeqKit (version 2.2.0) [30] is used as follows to create the reverse complement of all sequences because `bcftools consensus` outputs the sequence from the forward strand:

```
seqkit seq -r -p "${name}.fa" -o "${name}_rc.fa"
```

The variable `name` represents the name of the current miRNA. The creation of the reverse sequence is specified by `-r` and the creation of the complement sequence by `-p`. `$name.fa` is the input file and `-o` defines the output file.

Subsequently, the resulting FASTA files were used to identify the unique haplotype sequences for each miRNA. These haplotype sequences were aligned using the R [25] packages `msa` (version 1.42.0) [2] and `Biostings` (version 2.77.2) [24]. Variants of all haplotypes compared to the reference sequence were visualized per miRNA using heatmaps. The sequences with the highest frequencies were used as reference sequences. Moreover, the haplotype sequences were saved with their frequencies along with the multiple sequence alignments.

2.11 Structure prediction of pri-miRNAs

The minimum free energy (MFE) secondary structures of all unique haplotype sequences were predicted using RNAfold (version 2.7.2) [18]. The `-p` argument was passed to RNAfold to compute base pairing probabilities for subsequent structure coloring, and `--noLP` was applied to exclude lonely base pairs which are very unstable [18]. Subsequently, the positional entropy was incorporated into the secondary structures to evaluate whether regions are well defined [18]. For this purpose, the `relplot.pl` Perl script was used with the line for the legend commented out. To convert the MFE secondary structures to PDF files, ImageMagick (version 7.1.2-28) was used. This resulted in one PDF file per haplotype sequence. To generate a single PNG file per pri-miRNA containing all haplotype structures, R (version 4.5.1) [25] was utilized with the packages `magick` (version 2.9.1) [21] and `pdftools` (version 3.9.0) [22].

3 Results

This section presents the results of the analyses conducted in this thesis. To guide the reader, the following subsections are structured in the same order as in the previous section. All scripts, raw data, and output files are available via GitHub (<https://github.com/CajusHartmann/Bachelors-Thesis>).

3.1 Genotype heatmaps

In the course of this thesis, a total of 332 genotype heatmaps of the miRNA regions were created to examine the presence and distribution of variants. 237 of them belong to the predicted miRNAs and 95 to the non-predicted miRNAs. In contrast to this, the database miRBase (version 22) only contains entries for 326 pri-miRNAs. More than 326 heatmaps were created because *MIR161*, *MIR447a*, *MIR779*, *MIR780*, *MIR829*, *MIR869* and *MIR1886* have two miRNA/miRNA* duplexes annotated and therefore two heatmaps were generated. For *MIR156i* no heatmap was created because no prediction could be made (see Subsection 2.5). Moreover, 332 genotype heatmaps filtered for a MAF greater than or equal to 0.005 and a missing rate less than or equal to 0.5 were generated to exclude variant positions with limited data or variants.

The following text explains the genotype heatmaps in detail, using Figure 5 as an example. The rows show the 1135 accessions of *A. thaliana* and the columns represent the positions within the pri-miRNAs. The upper axis indicates the exact position, while the lower axis shows the respective region in the pri-miRNA. The background is colored depending on the region. Regions are always arranged such that miRNA-5p is on the left. White spots indicate missing data, light blue spots represent the reference base, and red spots indicate variants. When there is no bar in the plot, the position is not present in the VCF file, meaning no variant was detected. Figure 5 shows that *MIR156a* comprises variants within the base regions. Closer inspection reveals that the first variant position contains two variants (red), whereas the other accessions carry the reference base (light blue) or have no data available (white). At the second variant position all accessions lack data except for one with a variant. The other variant positions also feature only a few variants, except for the penultimate, which contains a higher number of variants.

The presence and distribution of variants varies greatly among *MIR* genes. This variability is shown in Figure 6 for non-predicted miRNAs and in Figure 7 for predicted miRNAs. A comparison between non-filtered and filtered heatmaps is presented in Figure 8.

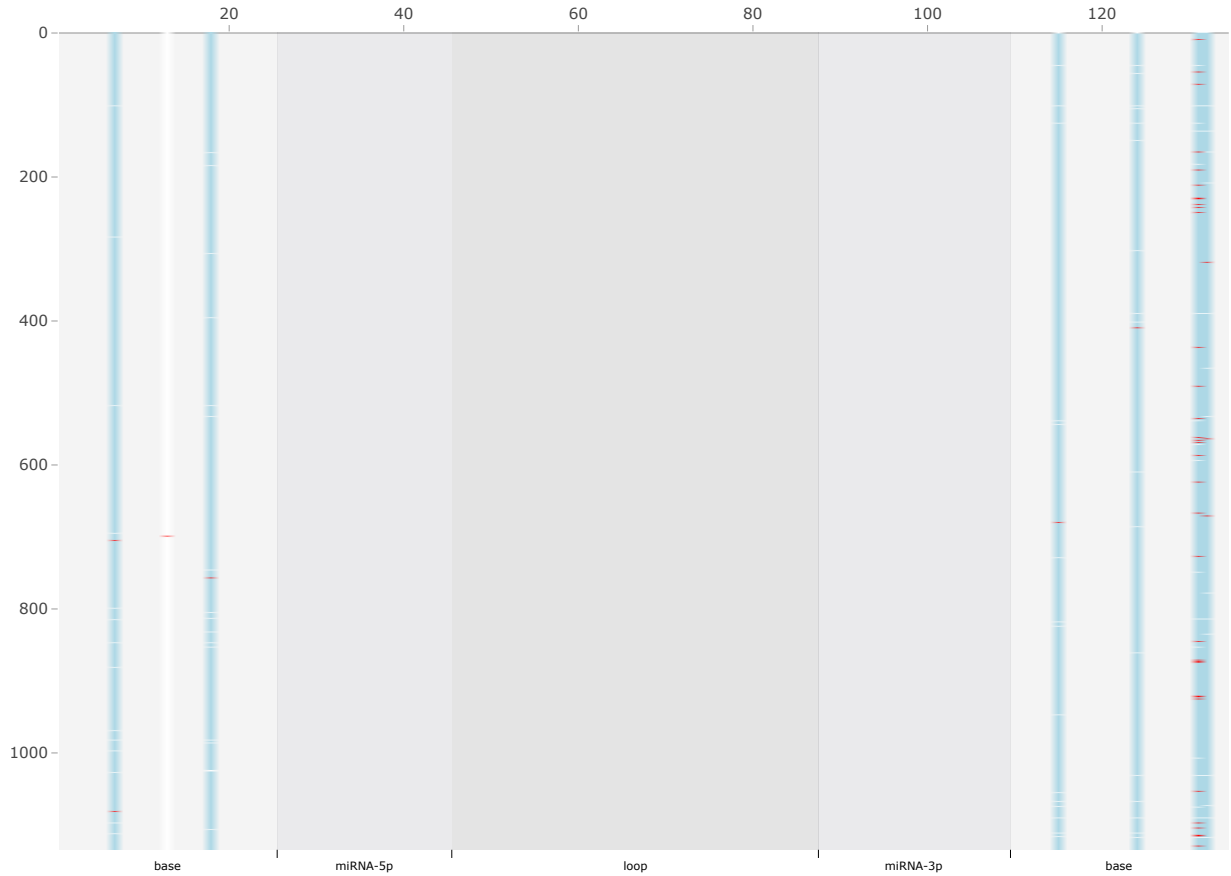


Figure 5: Genotype heatmap of *MIR156a*

MIR156a contains seven variant positions. Three are located in the base region near the 5' end and four in the base region near the 3' end of the pri-miRNA. The first variant position contains two variants (red), whereas the other accessions carry the reference base (light blue) or have no data available (white). At the second variant position most accessions lack data (white). The penultimate variant position comprises a higher number of variants.

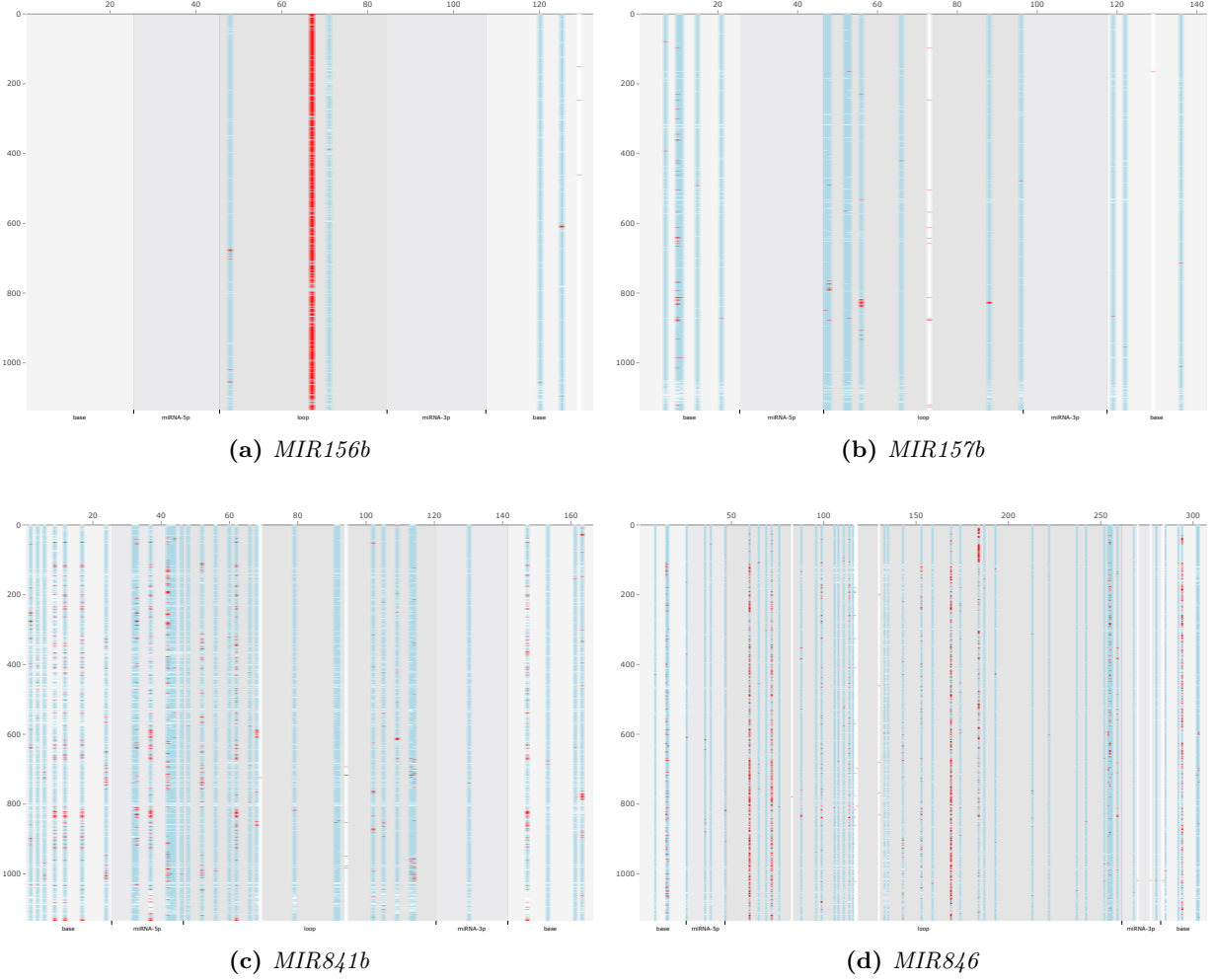


Figure 6: Genotype heatmaps of non-predicted miRNAs

Figure 6a contains a noticeable red bar in the loop region demonstrating a high frequency of variants compared to the reference at this position. However, these variants could all be identical, which would indicate a highly conserved position. Compared to Figure 6a, Figure 6b comprises more variant positions and displays a slightly longer loop. This trend continues in Figure 6c, which shows an even longer loop and a higher amount of variant positions. Figure 6d demonstrates the longest loop and highest number of variant positions of the presented non-predicted miRNAs.

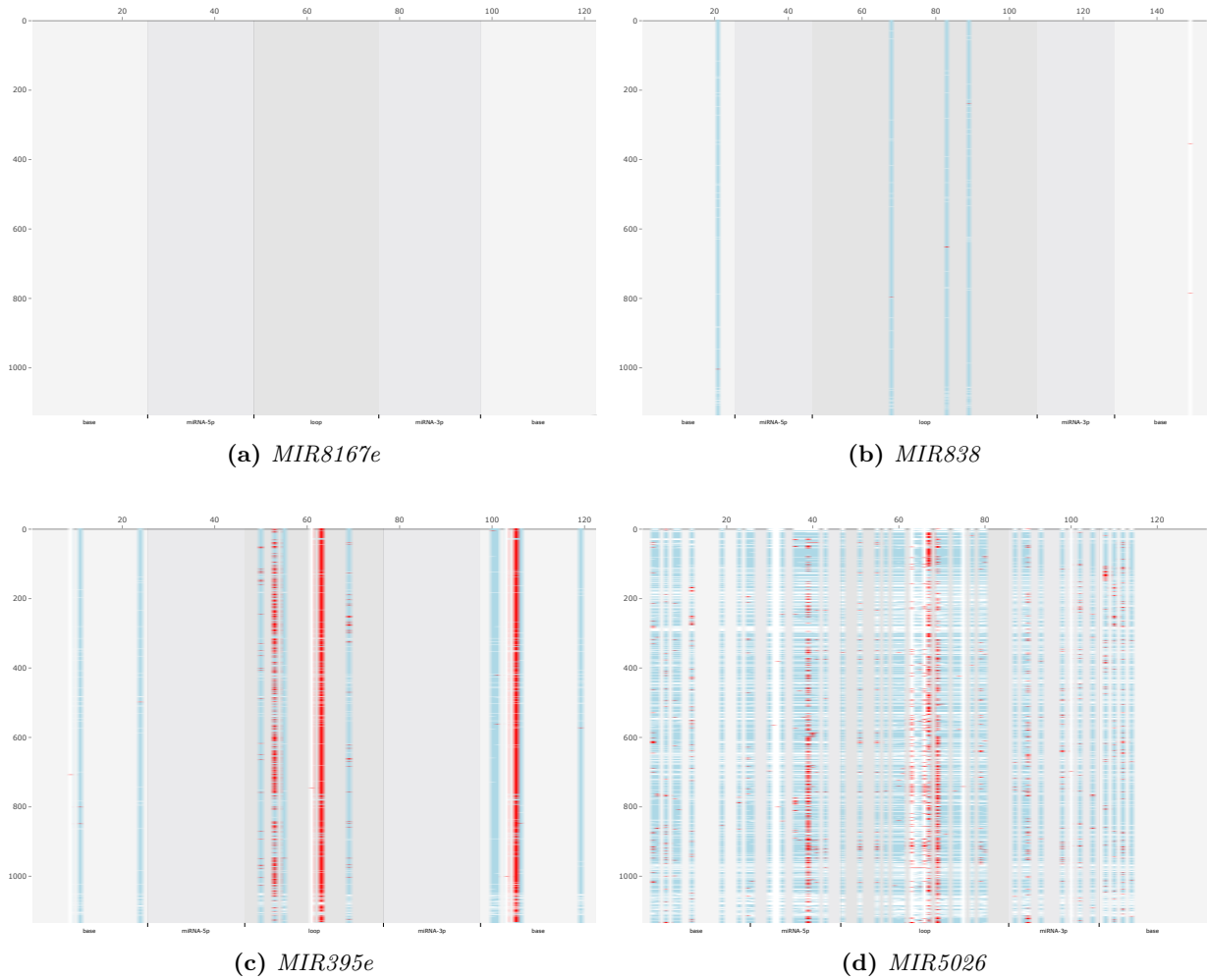


Figure 7: Genotype heatmaps of predicted miRNAs

Figure 7a contains no variants, whereas Figure 7b features a few variant positions, one of which consists mostly of missing data. The red bars in Figure 7c highlight positions with a high number of variants compared to the reference. However, these variants could all be identical, which would indicate highly conserved positions. Among all predicted miRNAs presented, Figure 7d displays the highest number of variant positions.

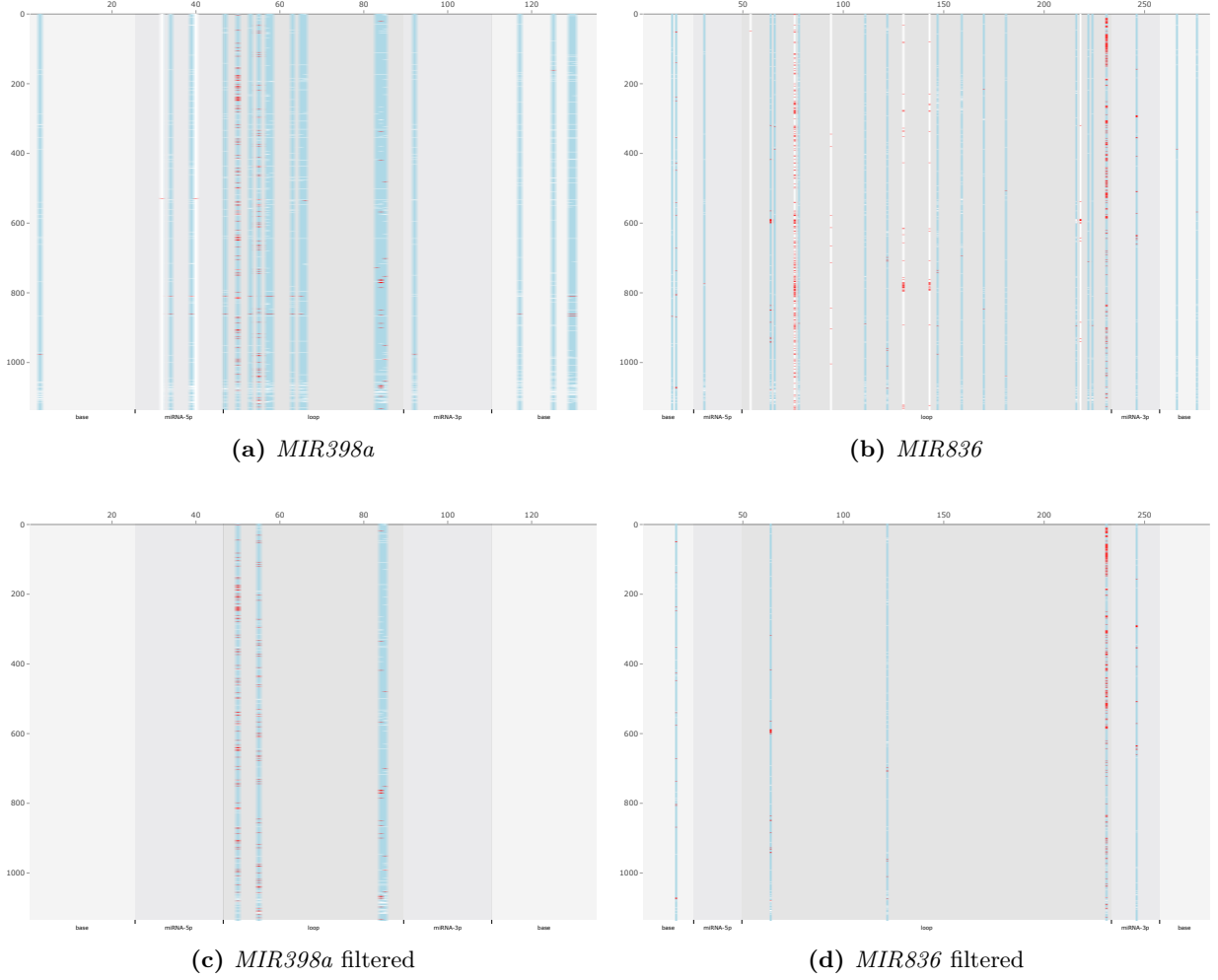


Figure 8: Comparison of non-filtered and filtered heatmaps

Compared to Figure 8a and Figure 8b, Figure 8c and Figure 8d display fewer variant positions because these genotype heatmaps were filtered for a MAF greater than or equal to 0.005 and a missing rate less than or equal to 0.5. In the miRNA-5p region of Figure 8a, the two white bars were filtered out due to their MAF and missing rate, while the two light blue bars were removed solely based on their MAF. In the loop region of Figure 8b, the second bar with a high frequency of missing data (white) was filtered exclusively because of its missing rate and not based on its MAF. *MIR398a* belongs to the non-predicted miRNAs and *MIR836* to the predicted miRNAs.

3.2 Plots per processing type

Table 2 shows the number of pri-miRNAs per processing type for the non-predicted and the predicted miRNAs as well as the total. All analyses described in this section were performed on the non-filtered data and separately for the non-predicted and the predicted miRNAs.

Table 2: Number of pri-miRNAs per processing type for the non-predicted and the predicted miRNAs as well as the total.

Processing type	Non-predicted	Predicted	Total
Bidirectional	28	15	43
BTL	13	26	39
LTB	17	22	39
SBTL	4	12	16
SLTB	4	10	14

3.2.1 Aggregated heatmaps

To get an overview of the heatmaps and to search for patterns across the processing types, aggregated heatmaps were created per processing type. A position in an aggregated heatmap reaches the maximum value of 1 (dark red) only if every heatmap of that processing type carries a variant there. The minimum value of 0 (dark blue) indicates that no heatmap carries a variant at that position. If data is missing across all heatmaps of a processing type at a position, no value is displayed. Values between 0 and 1 represent the proportion of heatmaps carrying a variant (e.g., 0.5 (gray) for half of the heatmaps).

Figure 9 presents the aggregated heatmaps for the non-predicted miRNAs. For all processing types the displayed values are close to 0 (dark blue), indicating that variants are not positionally consistent across multiple pri-miRNAs within a processing type. However, differences in the number of variant positions are visible between the processing types. Those differences are probably a consequence of the varying group sizes (see Table 2).

Figure 10 demonstrates the aggregated heatmaps for the predicted miRNAs and reveals a pattern similar to that in Figure 9. Again, all displayed values are near 0 (dark blue) and the observed differences in the number of variant positions can probably also be attributed to the varying group sizes.

RESULTS

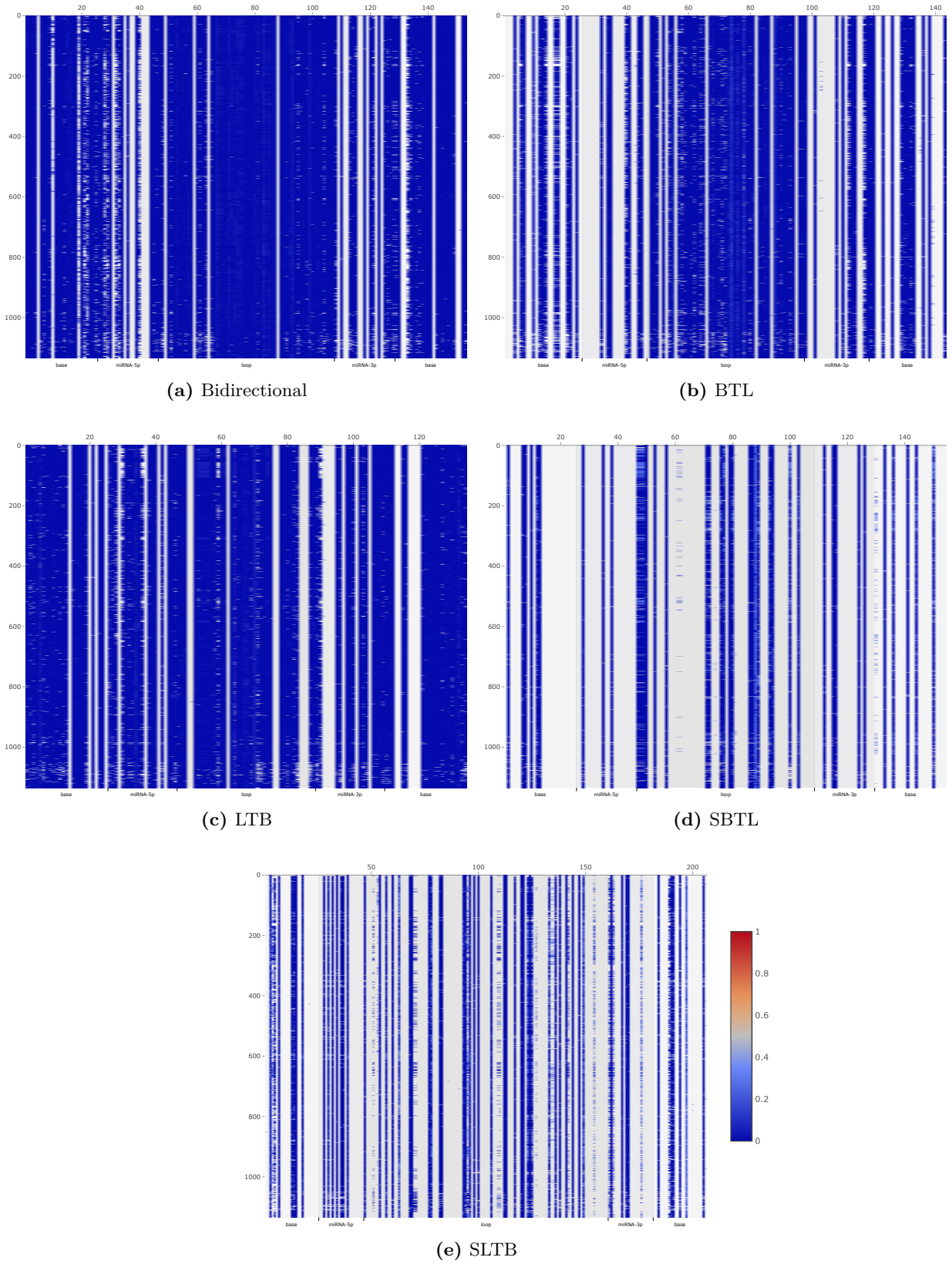


Figure 9: Aggregated heatmaps of the non-predicted miRNAs

All aggregated heatmaps display values close to 0 (blue), indicating that variants at exactly the same positions do not occur in the majority of heatmaps within a group. The loops differ in length and the heatmaps show varying densities of variant positions. The bidirectional and LTB groups are the most densely populated, followed by the BTL, SLTB, and SBTL groups.

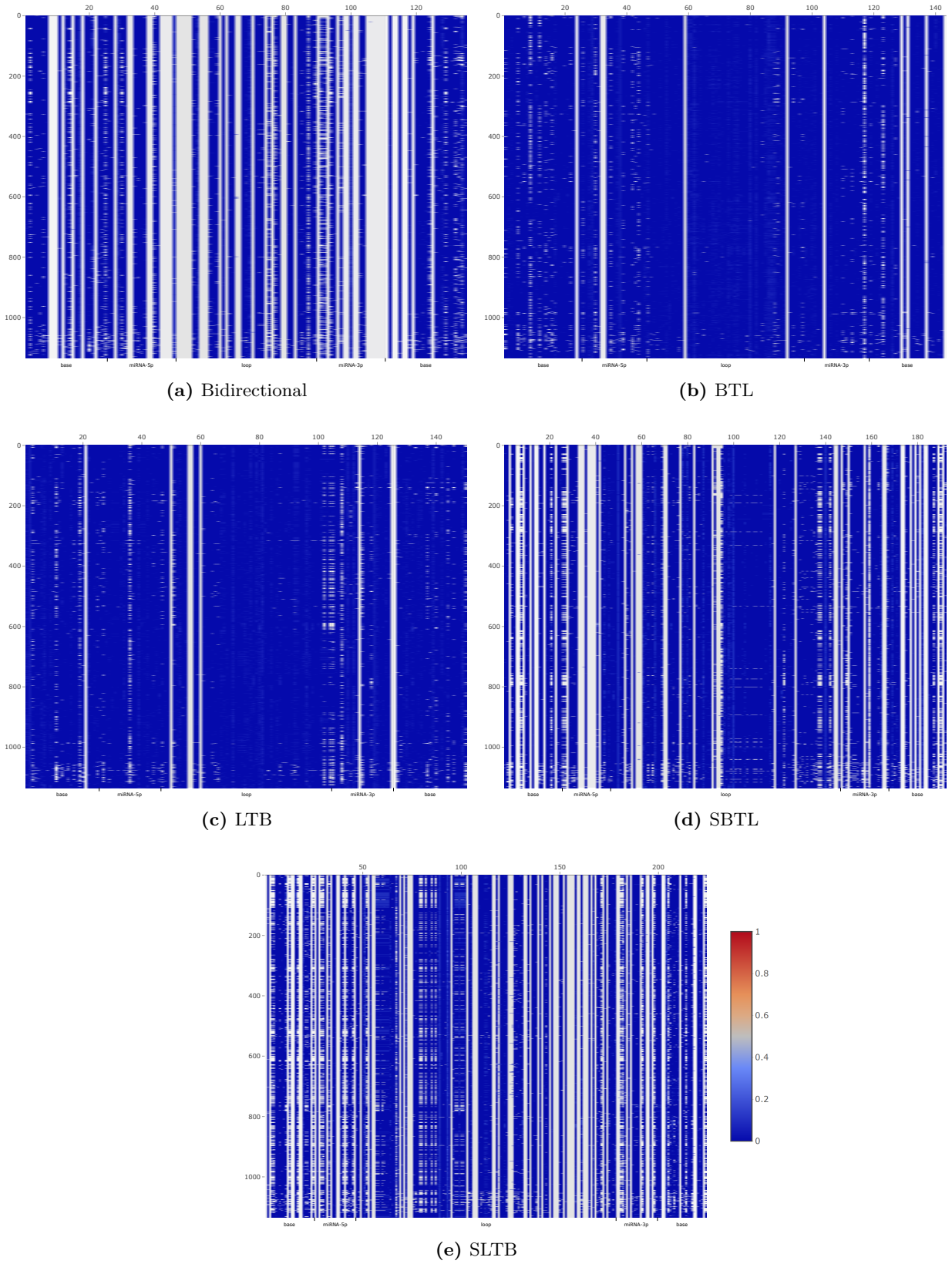


Figure 10: Aggregated heatmaps of the predicted miRNAs

All aggregated heatmaps display values close to 0 (blue), indicating that variants at exactly the same positions do not occur in the majority of heatmaps within a group. The loops differ in length and the heatmaps show varying densities of variant positions. The LTB and BTL groups are the most densely populated, followed by the SBTL, SLTB, and bidirectional groups.

3.2.2 Variant frequency plots

To further search for patterns across the processing types, variant frequency plots were generated that show the variant frequency across the 1135 accessions on the y-axis and the position in the pri-miRNA on the x-axis. The miRNA-5p is always the one on the left side. The darkness of the lines indicates the number of pri-miRNAs of the processing type in which the variant occurs. Compared to the aggregated heatmaps, the advantage of the variant frequency plots is that high variant frequencies are directly visible, as the values are not normalized to the group size.

Figure 11 demonstrates the variant frequency plots for the non-predicted miRNAs. High variant frequencies represented by light blue lines – indicating that they occur in only one pri-miRNA – are directly visible in all subfigures but with different densities. The bidirectional group features the highest number of variants, followed by the BTL, LTB, SLTB, and SBTL groups. This observation could again be a consequence of the varying group sizes (see Table 2). Furthermore, all subfigures show a dark blue line at a frequency of 0 because many pri-miRNAs in each group contain numerous positions with few or no variants. In general, it appears that the miRNA regions are less affected by variants compared to the other regions although the miRNA-5p in the LTB group comprises high-frequency variants.

Figure 12 presents the variant frequency plots for the predicted miRNAs. Similar to Figure 11, high variant frequencies represented by light blue lines are directly visible in all subfigures with different densities and all subfigures show a dark blue line at a frequency of 0. The highest number of variants occurs in the LTB group, followed by the BTL, SLTB, SBTL, and bidirectional groups. Again, this can be caused by the varying group sizes. Compared to the non-predicted miRNAs, a higher proportion of variants is observed within the miRNA regions.

Even though high variant frequencies appear promising at first glance, these variants could all be identical which would consequently indicate highly conserved positions. Nevertheless, they could also be distinct resulting in high sequence diversity.

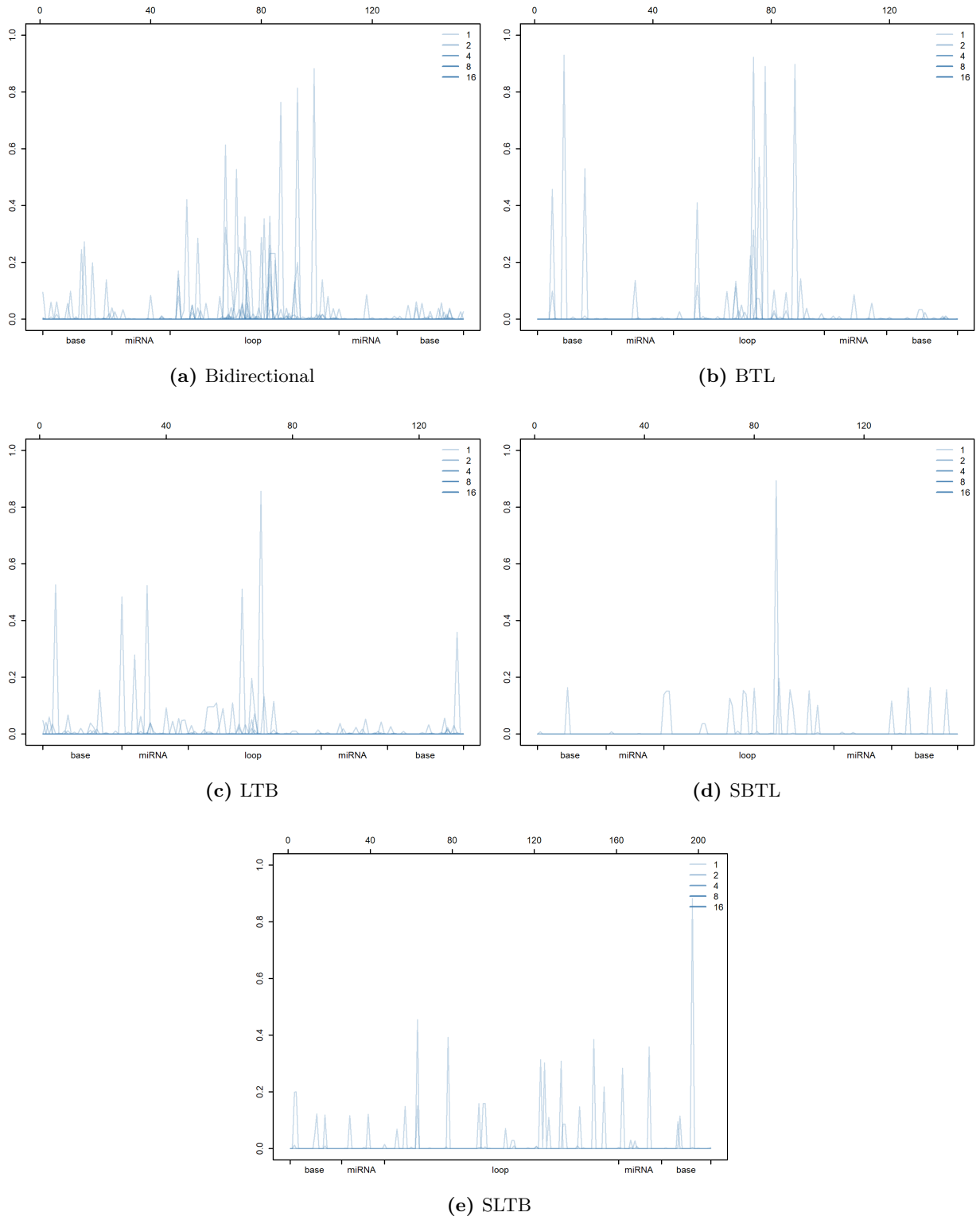


Figure 11: Variant frequency plots of the non-predicted miRNAs

All subfigures contain prominent high-frequency variants and a dark blue line at a frequency of 0 because many pri-miRNAs contain numerous positions without variants. The bidirectional group features the highest number of variants, followed by the BTL, LTB, SLTB, and SBTl groups. The loop regions differ in length.

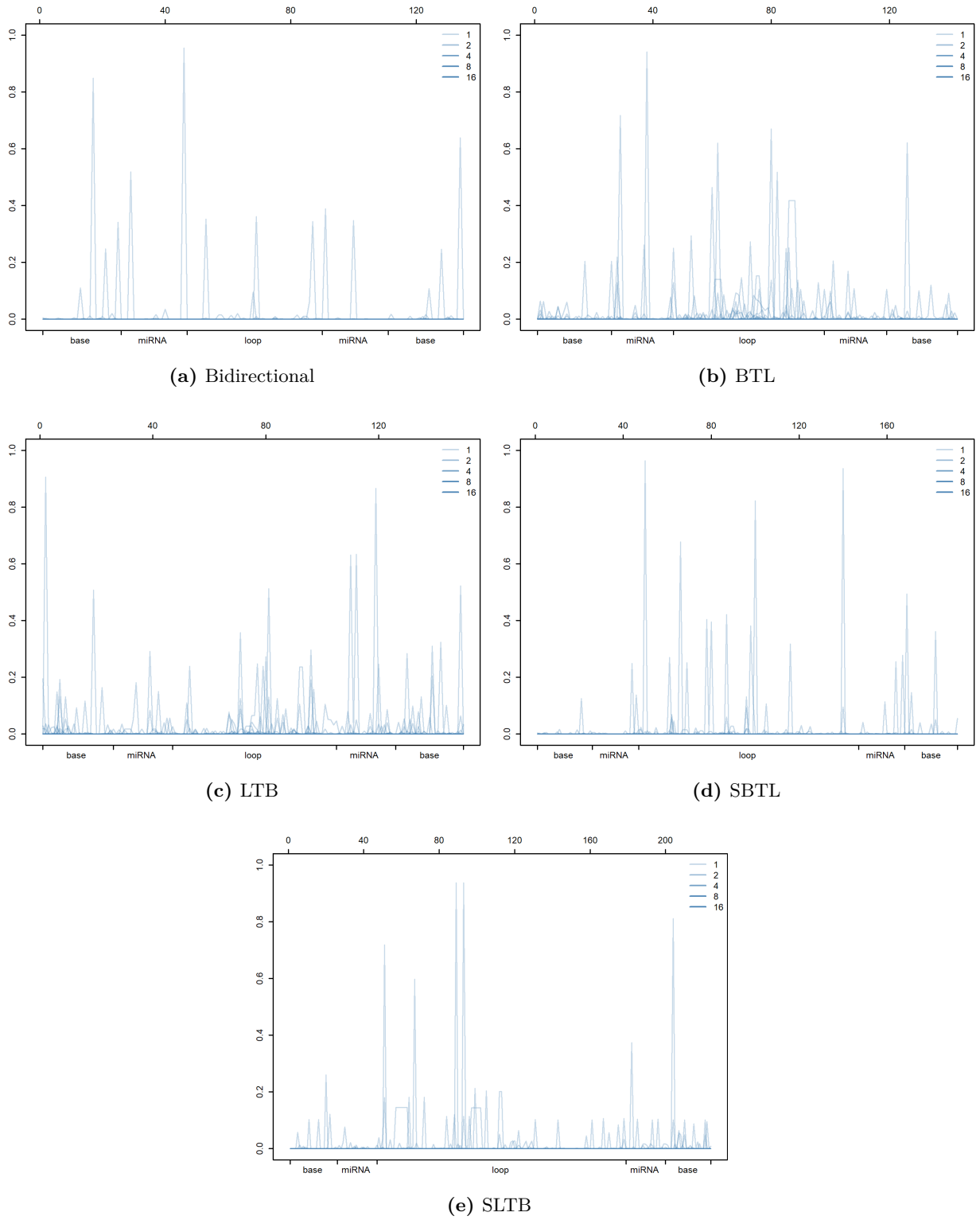


Figure 12: Variant frequency plots of the predicted miRNAs

All subfigures contain prominent high-frequency variants and a dark blue line at a frequency of 0 because many pri-miRNAs contain numerous positions without variants. The highest number of variants occurs in the LTB group, followed by the BTL, SLTB, SBTl, and bidirectional groups. The loop regions differ in length.

3.3 Statistical analyses

All analyses described in this section were performed on the non-filtered data and separately for the non-predicted and the predicted miRNAs.

3.3.1 Permutation test

To examine whether variants are more abundant in pri-miRNA regions compared to the whole genome, permutation tests were applied resulting in comma-separated files containing the names, p -values, z -scores, and alternative hypotheses of the miRNAs. Table 3 shows the results for the non-predicted miRNAs filtered by a p -value of 0.05, while Table 4 shows the corresponding results for the predicted miRNAs. Both tables are sorted in ascending order by the z -score.

After filtering, only two non-predicted miRNAs remained, whereas 38 predicted miRNAs were retained. The z -score quantifies the strength and direction of the association meaning that a high positive z -score indicates a substantially higher abundance of variants in miRNA regions than expected by chance, whereas a negative z -score indicates a reduced abundance.

To illustrate this, miR841a, miR854e, and miR8174 were selected as examples. Based on its z -score, miR841a is expected to contain a high number of variant positions which is confirmed by the heatmap (see Figure 13a). Conversely, miR854e should feature very few variant positions whereas miR8174 is expected to show a high number. These hypotheses are also supported by the corresponding heatmaps (see Figure 13b and Figure 13c).

Table 3: Filtered results of the permutation test for the non-predicted miRNAs

Name	p -value	z -score	Test
miR773b	0.033	2.57	greater
miR841a	0.016	2.84	greater

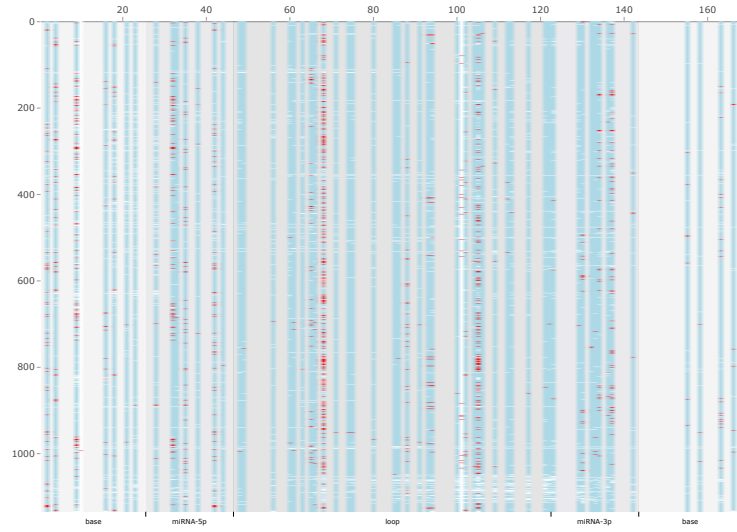
3.3.2 Kruskal-Wallis test

Differences between the processing types regarding the in the regions of the pri-miRNAs are shown in Table 5 for the non-predicted miRNAs and in Table 6 for the predicted miRNAs. To generate these tables, variants within each region were summed across all pri-miRNAs belonging to the same processing type and subsequently divided by the lengths of the regions. Consequently, absolute numbers cannot be directly compared across processing types, whereas percentages allow for a meaningful comparison.

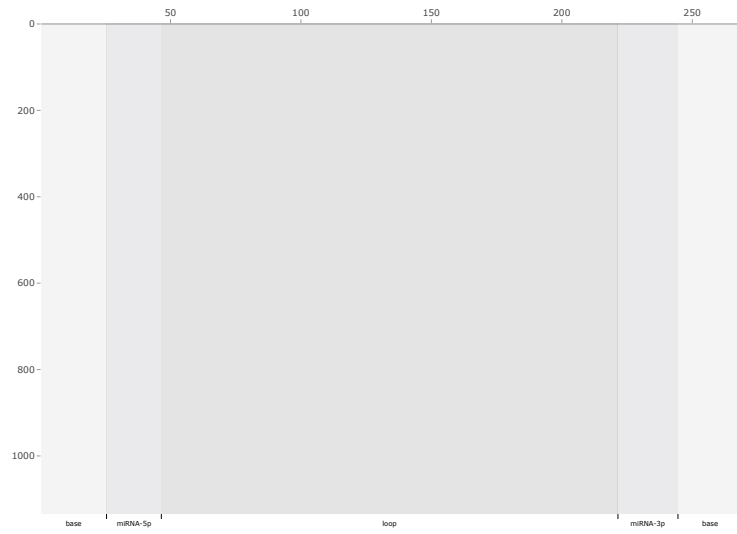
Particularly noticeable in Table 5 is the high proportion of variants in the miRNA-5p region of the LTB group, as well as in the miRNA regions of the SLTB group. Moreover, the loop region of the bidirectional group features a high percentage of variants. In Table 6, the miRNA regions appear less conserved and differences in the percentage of variants within the loop regions can be observed across the processing types. To assess whether these observed differences are statistically significant, a Kruskal-Wallis test was performed. This test revealed no statistically significant differences across the processing types for any of the regions, regardless whether the non-predicted or predicted miRNAs were analyzed, or whether the filtered data (see subsection 2.9.2) was utilized.

Table 4: Filtered results of the permutation test for the predicted miRNAs

Name	p -value	z -score	Test
miR854e	0.025	-1.50	less
miR854d	0.022	-1.46	less
miR8167e	0.029	-1.43	less
miR8167d	0.021	-1.42	less
miR854b	0.019	-1.42	less
miR854c	0.039	-1.41	less
miR5642b	0.023	-1.37	less
miR5642a	0.011	-1.36	less
miR5630b	0.046	-1.34	less
miR854a	0.035	-1.31	less
miR8167b	0.026	-1.23	less
miR5645e	0.042	2.06	greater
miR5645b	0.045	2.14	greater
miR405a	0.047	2.18	greater
miR5645a	0.035	2.48	greater
miR1888b	0.034	2.48	greater
miR5666	0.035	2.49	greater
miR5016	0.022	2.54	greater
miR5020b	0.023	2.66	greater
miR4245	0.026	2.67	greater
miR5628	0.024	2.71	greater
miR8177	0.023	2.71	greater
miR416	0.021	2.75	greater
miR5645c	0.022	2.82	greater
miR5664	0.020	2.84	greater
miR776	0.021	2.86	greater
miR5644	0.022	2.90	greater
miR5659	0.020	3.00	greater
miR5645d	0.016	3.16	greater
miR5662	0.006	3.25	greater
miR406	0.010	3.60	greater
miR405b	0.003	3.61	greater
miR8168	0.006	3.27	greater
miR8180	0.011	3.27	greater
miR1888a	0.005	3.46	greater
miR5026	0.005	3.49	greater
miR8174	0.003	4.45	greater
miR8166	0.002	4.65	greater



(a) *MIR841a*



(b) *MIR854e*



(c) *MIR8174*

Figure 13: Genotype heatmaps supporting the results of the permutation test

Table 5: Number of variants per nucleotide in the regions of the non-predicted miRNAs per processing type. Corresponding percentages indicate the relative distribution of variants across the pri-miRNA regions for each processing type (row totals are 100%). To calculate the number of variants per nucleotide, the total number per region and processing type was divided by the length of the region. For the loop the respective median length was used (see Table 1). Bases and miRNAs were divided by 25 and 21.

Processing type	base	miRNA-5p	loop	miRNA-3p	base
Bidirectional	62 (17.97%)	13 (3.77%)	233 (67.54%)	9 (2.61%)	28 (8.12%)
BTL	93 (37.65%)	9 (3.64%)	128 (51.82%)	10 (4.05%)	7 (2.83%)
LTB	52 (19.40%)	96 (35.82%)	78 (29.10%)	13 (4.85%)	29 (10.82%)
SBTL	8 (8.70%)	1 (1.09%)	54 (58.70%)	1 (1.09%)	28 (30.43%)
SLTB	33 (19.08%)	14 (8.09%)	36 (20.81%)	39 (22.54%)	51 (29.48%)

Table 6: Number of variants per nucleotide in the regions of the predicted miRNAs per processing type. Corresponding percentages indicate the relative distribution of variants across the pri-miRNA regions for each processing type (row totals are 100%). To calculate the number of variants per nucleotide, the total number per region and processing type was divided by the length of the region. For the loop the respective median length was used (see Table 1). Bases and miRNAs were divided by 25 and 21.

Processing type	base	miRNA-5p	loop	miRNA-3p	base
Bidirectional	74 (26.06%)	83 (29.23%)	36 (12.68%)	41 (14.44%)	50 (17.61%)
BTL	39 (8.04%)	149 (30.72%)	174 (35.88%)	57 (11.75%)	66 (13.61%)
LTB	145 (24.13%)	64 (10.65%)	107 (17.80%)	165 (27.45%)	120 (19.97%)
SBTL	10 (4.69%)	24 (11.27%)	82 (38.50%)	40 (18.78%)	57 (26.76%)
SLTB	32 (14.41%)	10 (4.50%)	57 (25.68%)	45 (20.27%)	78 (35.14%)

3.4 Haplotype sequences

For all pri-miRNAs for which a genotype heatmap was generated, another heatmap visualizing the haplotype sequences was created. That means that a total of 332 haplotype heatmaps (95 for the non-predicted and 237 for the predicted miRNAs) was created. Furthermore, for every pri-miRNA the DNA sequence and the corresponding frequencies of the haplotypes among the 1135 accessions were saved together with the multiple sequence alignments. Only variants with a MAF of at least 1% were included in these analyses.

The following text explains the haplotype heatmaps in detail, using Figure 14 as an example. The rows represent the different haplotypes with their frequencies and the columns show the positions within the pri-miRNAs. When multiple haplotypes have identical frequencies, a suffix is appended to establish a unique identifier (e.g., 1x and 1x.1 for two haplotypes with a frequency of one). The upper axis indicates the exact position, whereas the lower axis shows the respective region in the pri-miRNA. Regions are always arranged such that miRNA-5p is on the left. Red spots in the heatmaps indicate variants. The top sequence is always the one with the highest frequency and serves as the reference, meaning that it never contains variants. Figure 14 shows that four haplotypes of *MIR159b* exist among the 1135 accessions. The haplotype with a frequency of 11 features one variant in the base close to the 5' end, whereas the haplotype with a frequency of 8 contains a variant within the loop. The haplotype observed four times comprises one variant in the 5' base and one in the loop. It should be noted that indels longer than one nucleotide are not displayed as a single variant, but as a thicker bar depending on the indel size, because the haplotype heatmaps are based on the multiple sequence alignments. This differs from the genotype heatmaps.

The number of haplotype sequences varies greatly among *MIR* genes. Some *MIR* genes only feature one haplotype such as *MIR160a* and others comprise more than 100 haplotypes such as *MIR824* with 303. Moreover, the presence and distribution of variants differs widely among haplotypes and *MIR* genes. This variability is shown in Figure 15.

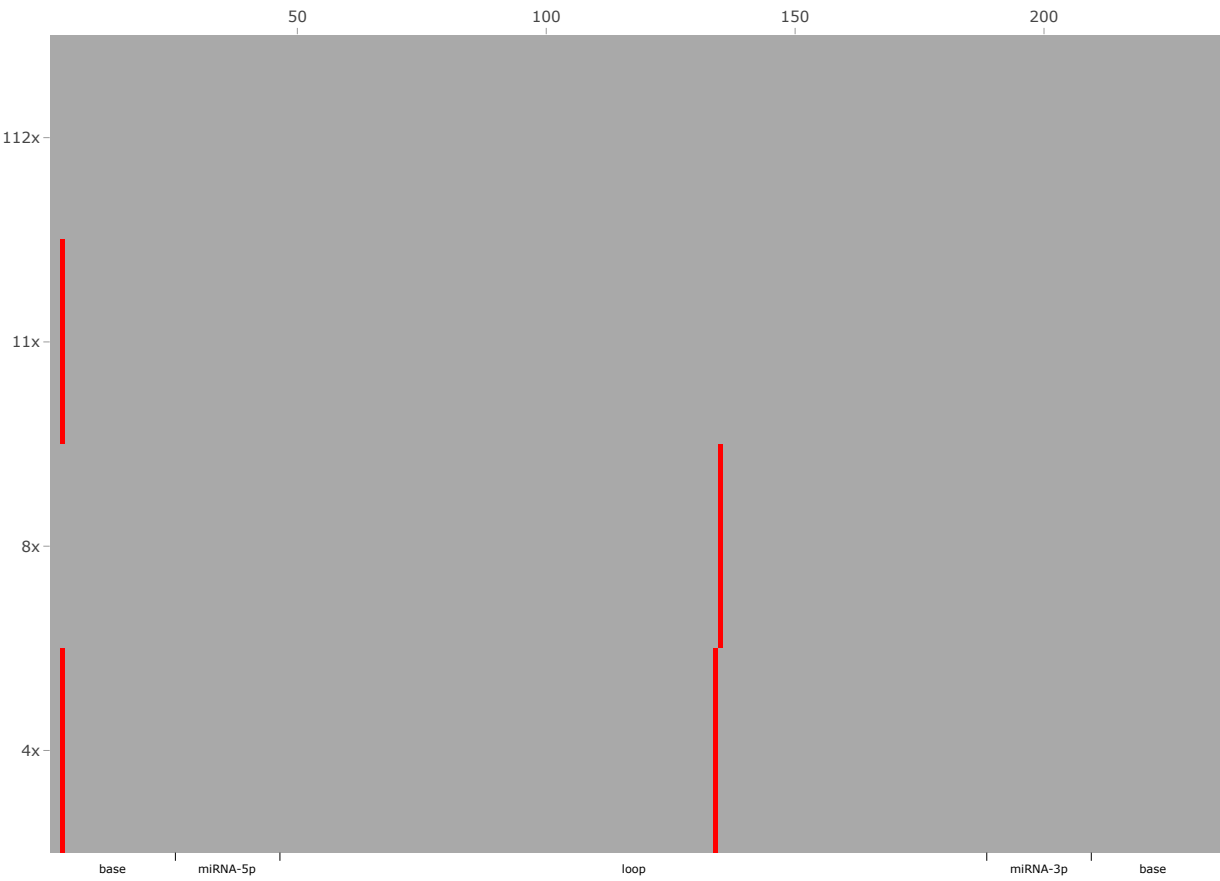
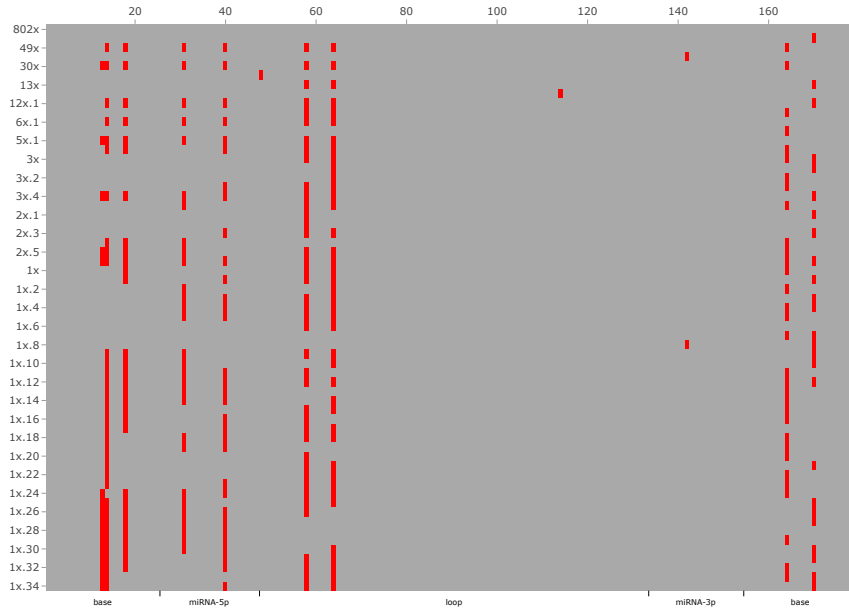
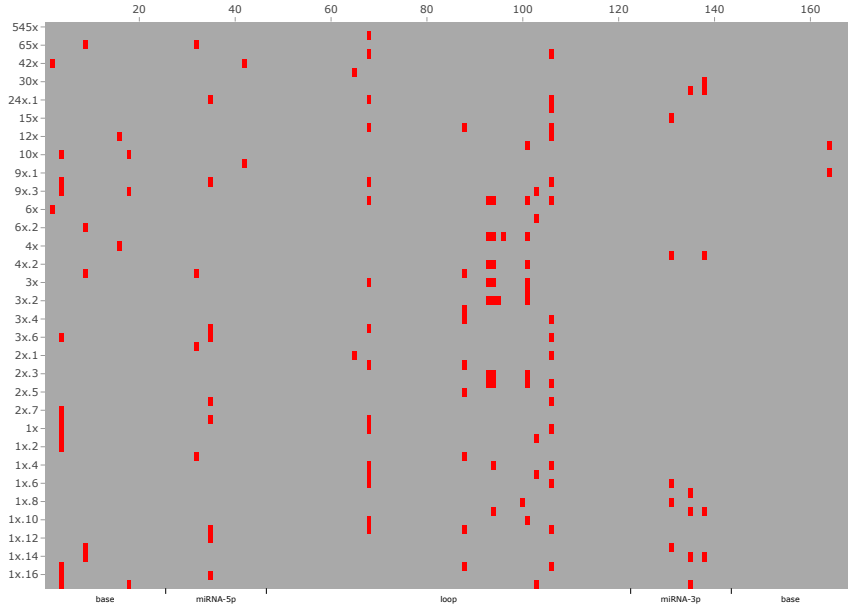


Figure 14: Haplotype heatmap of *MIR159b*
Four haplotypes of *MIR159b* exist among the 1135 accessions. Variants (red) occur in the 5' base and in the loop.

(a) *MIR840*(b) *MIR841a***Figure 15:** Haplotype heatmaps of *MIR840* and *MIR841a*

MIR840 features 61 haplotypes and *MIR841a* 62. The presence and distribution of variants varies widely among haplotypes which can be seen in Figure 15a by comparing haplotype 3x.4 with haplotype 1x.8. Haplotype 3x.4 comprises variants within every region of the pri-miRNA, whereas haplotype 1x.8 only contains variants in the miRNA-3p region and in the 3' base. Furthermore, the distribution of variants differs greatly among the haplotypes of different *MIR* genes, as supported by comparing Figure 15a with Figure 15b. Figure 15a displays variants at only 12 positions but these variants occur in a lot of haplotypes, whereas Figure 15b shows variants at 23 positions that mostly occur in only a few haplotypes.

3.5 Structure prediction of pri-miRNAs

To visualize the effects of the variants on the pri-miRNA secondary structures, the MFE secondary structures of all haplotype sequences were predicted. One PNG file was generated per pri-miRNA (except for pri-miR824 due to its 303 haplotypes) containing all haplotype structures. Figure 16 shows such a file for pri-miR156a with the corresponding legend for the haplotype occurring 1001 times. The frequencies of the haplotypes are shown above the structures, but in general, haplotypes are not sorted based on their frequency. The values of the legend vary between different structures, but the colors encode the same information, namely the level of entropy [12]. Therefore, in the following legends are not shown. Low entropy is displayed in red and increases via yellow and green to blue and violet for regions with high entropy [12]. A low entropy is associated with well defined regions and a high entropy with poorly defined regions [12]. In the following representations, the base of the pri-miRNAs is always located at the top, while the loop is located at the bottom.

The structures of two out of the nine haplotypes of the BTL-processed pri-miR173 are presented in Figure 17. The structure on the left occurs with a frequency of 19 and the one on the right with a frequency of 39. In both structures, the miRNA-3p is located on the right strand, directly below the internal loop positioned slightly above the middle of the structures, with the sequence 3'-AAAGCGAAUGUGUCUUAGU-5'. The two haplotypes show structural differences towards the base caused by a variant located 12 nt from the 3' end of the miRNA-3p towards the base. The haplotype on the left features the sequence 5'-CU-3', whereas the right one contains 5'-UC-3'. Consequently, the haplotypes exhibit structural differences in the region 11–16 nt from the miRNA-3p towards the base. The loop varies between the haplotypes because of a SNP from U to G at position 79 counting from the 5' end (upper left end) in the haplotype occurring 19 times.

In Figure 18 the structures of two out of the 62 haplotypes of the LTB-processed pri-miR841a are presented. The structure on the left occurs with a frequency of 11 and the one on the right with a frequency of 12. In both structures, the miRNA-5p is located on the left strand, directly above the fourth internal loop from the top of the left haplotype, having the sequence 5'-UACGAGCCACUUGAAACUGAA-3'. The left haplotype features one internal loop less in the base than the right one due to a SNP from U to A at position 16 (counting from the 5' end). Moreover, structural differences begin 12 nt from the 3' end of the miRNA-5p towards the loop because of SNPs and an indel.

The structures of two out of the four haplotypes of the bidirectionally-processed pri-miR391 are presented in Figure 19. In both structures, the miRNA-5p is located on the left strand and starts at the third internal loop from the top with the sequence 5'-UUCGCAGGAGAGAUAGCGCCA-3'. The only variant is a SNP from U to C in the left haplotype at position 15 from the 5' end. This

causes the first internal loop, positioned 9 nt from the 5' end of the miRNA-5p towards the base, to be significantly bigger.

In Figure 20 the structures of three out of the five haplotypes of the bidirectionally-processed pri-miR394b are presented. Terminal branched loops, which represent a key motif of bidirectionally processed pri-miRNAs (see Subsection 1.2), can be observed in the three structures. Furthermore, the miRNA-5p is consistently located on the left strand, directly under the third internal loop from the top, with the sequence 5'-UUGGCAUUCUGUCCACCUC-3'. The only variant occurs at position 16 from the 5' end, where A is substituted with G in the haplotype with a frequency of 295, and with C in the haplotype occurring only once. Compared to the haplotype observed with a frequency of 705, the structure of the middle haplotype remains identical. In contrast, the SNP in the left structure leads to a larger internal loop located 9 nt from the 5' end of the miRNA-5p towards the base.

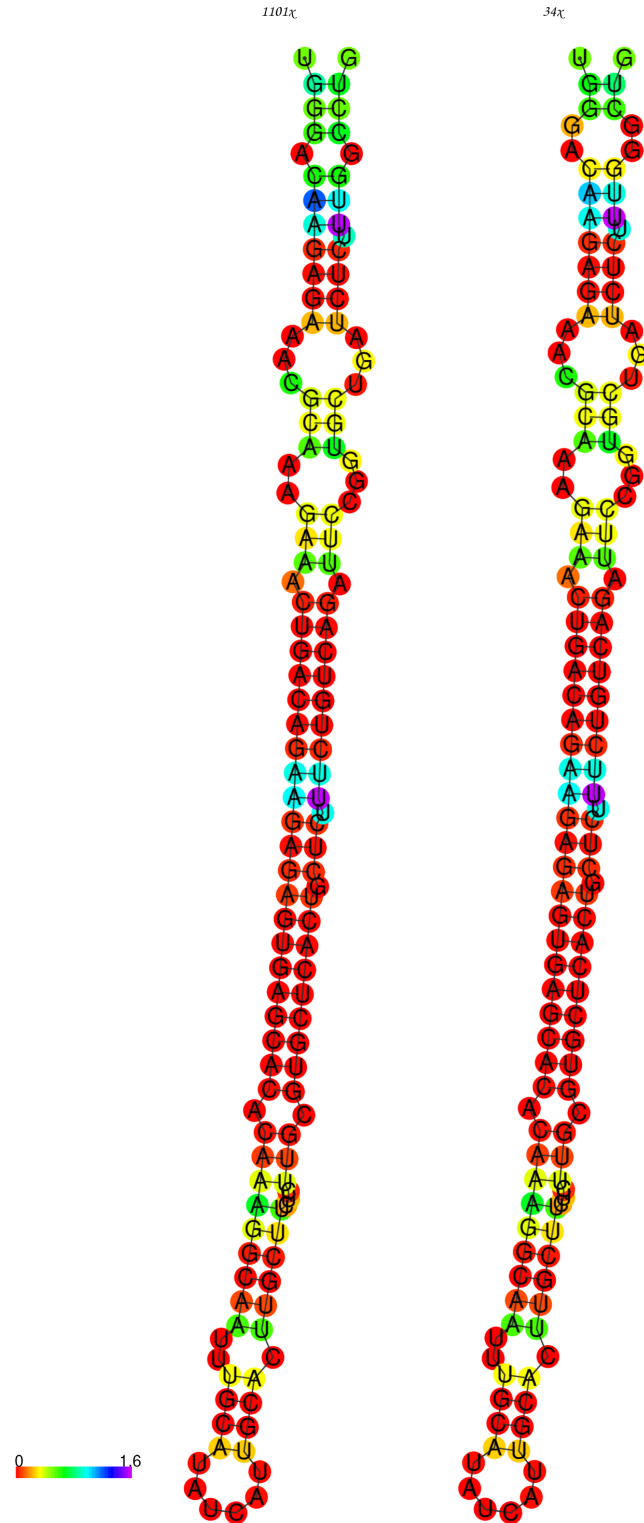


Figure 16: MFE secondary structure of pri-miR156a

Due to a SNP from C to G in the haplotype occurring 34 times, a larger bulge appears at the start of the stem of the pri-miRNA. Except for this region, the whole structure is identical, including the entropies. Directly under the fourth internal loop, counting from the top, the miRNA-5p starts on the left strand with the sequence 5'-UGACAGAAGAGAGUGAGCAC-3'. The miRNA-3p is located across the miRNA-5p with two nucleotide overhangs at both 3' ends of the miRNAs, resulting in the fourth internal loop, counting from the top, being the 3' end of the miRNA-3p. Entropies of nearly all of the bases belonging to the miRNA regions are very low, as indicated by red, showing a well defined region.

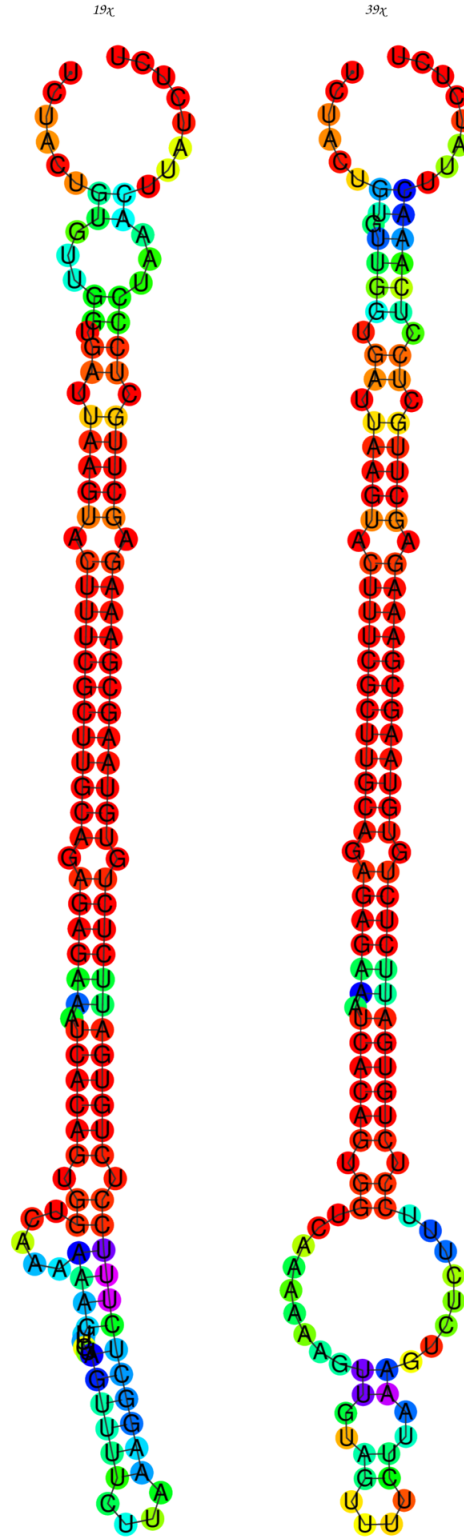


Figure 17: Secondary structures of two haplotypes of the BTL-processed pri-miR173

The base of the pri-miRNA is located at the top and the loop at the bottom. In both structures, the miRNA-3p is located on the right strand, directly below the internal loop positioned slightly above the middle of the structures, with the sequence 3'-AAAGCGAAUGUGUCUUAAGU-5'. Structural differences towards the base are caused by a variant located 12 nt from the miRNA-3p towards the base. The loop varies between the haplotypes because of a SNP from U to G at position 79 in the haplotype occurring 19 times.

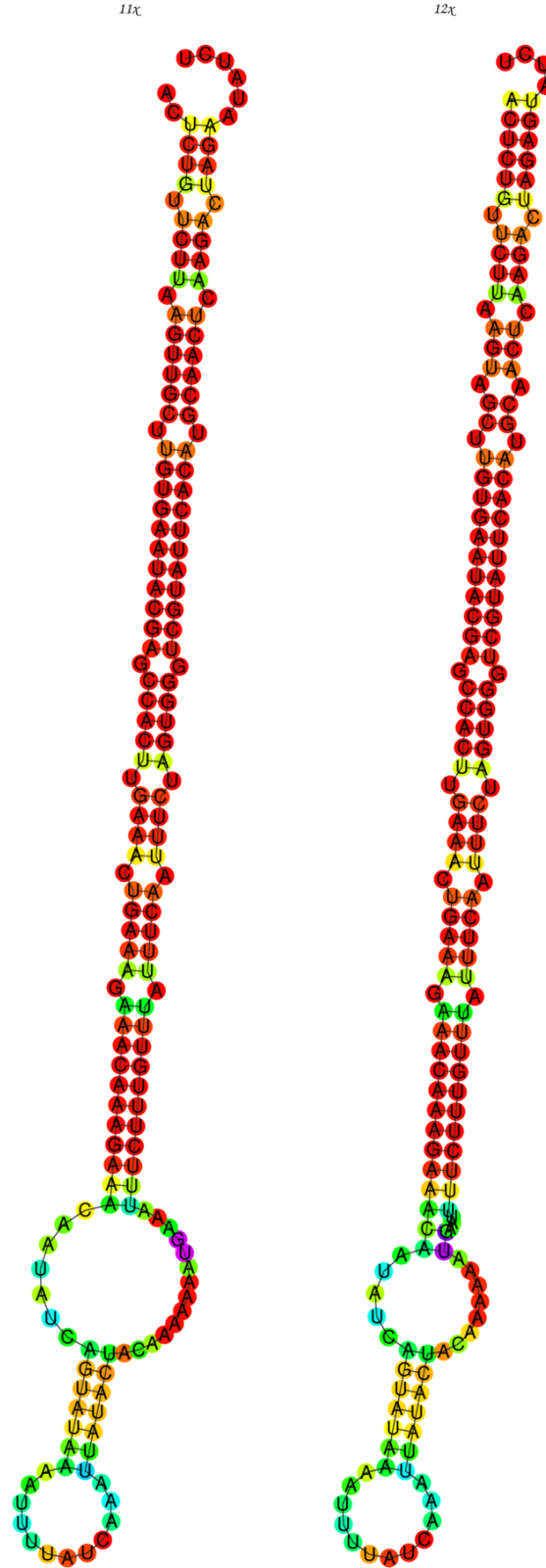


Figure 18: Secondary structures of two haplotypes of the LTB-processed pri-miR841a

The base of the pri-miRNA is located at the top and the loop at the bottom. In both structures, the miRNA-5p is located on the left strand, directly above the fourth internal loop (counting from the top of the left haplotype), with the sequence 5'-UACGAGCCACUUGAAACUGAA-3'. The left haplotype contains one internal loop fewer in the base due to a SNP and structural differences in the loops begin 12 nt from the 3' end of the miRNA-5p towards the loop because of SNPs and an indel.

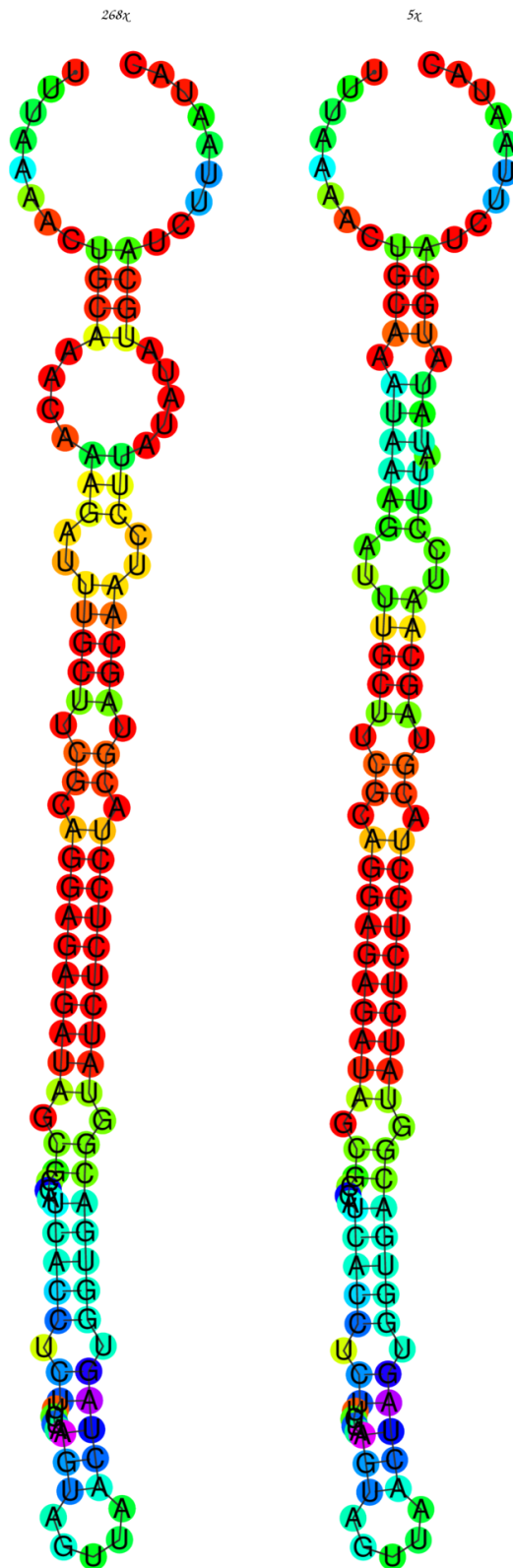


Figure 19: Secondary structures of two haplotypes of the bidirectionally-processed pri-miR391. The base of the pri-miRNA is located at the top and the loop at the bottom. In both structures, the miRNA-5p is located on the left strand and starts at the third internal loop from the top with the sequence 5'-UUCGCAGGAGAGAUAGCGCCA-3'. The first internal loop is positioned 9 nt from the 5' end of the miRNA-5p towards the base and is significantly bigger in the left haplotype due to a SNP.

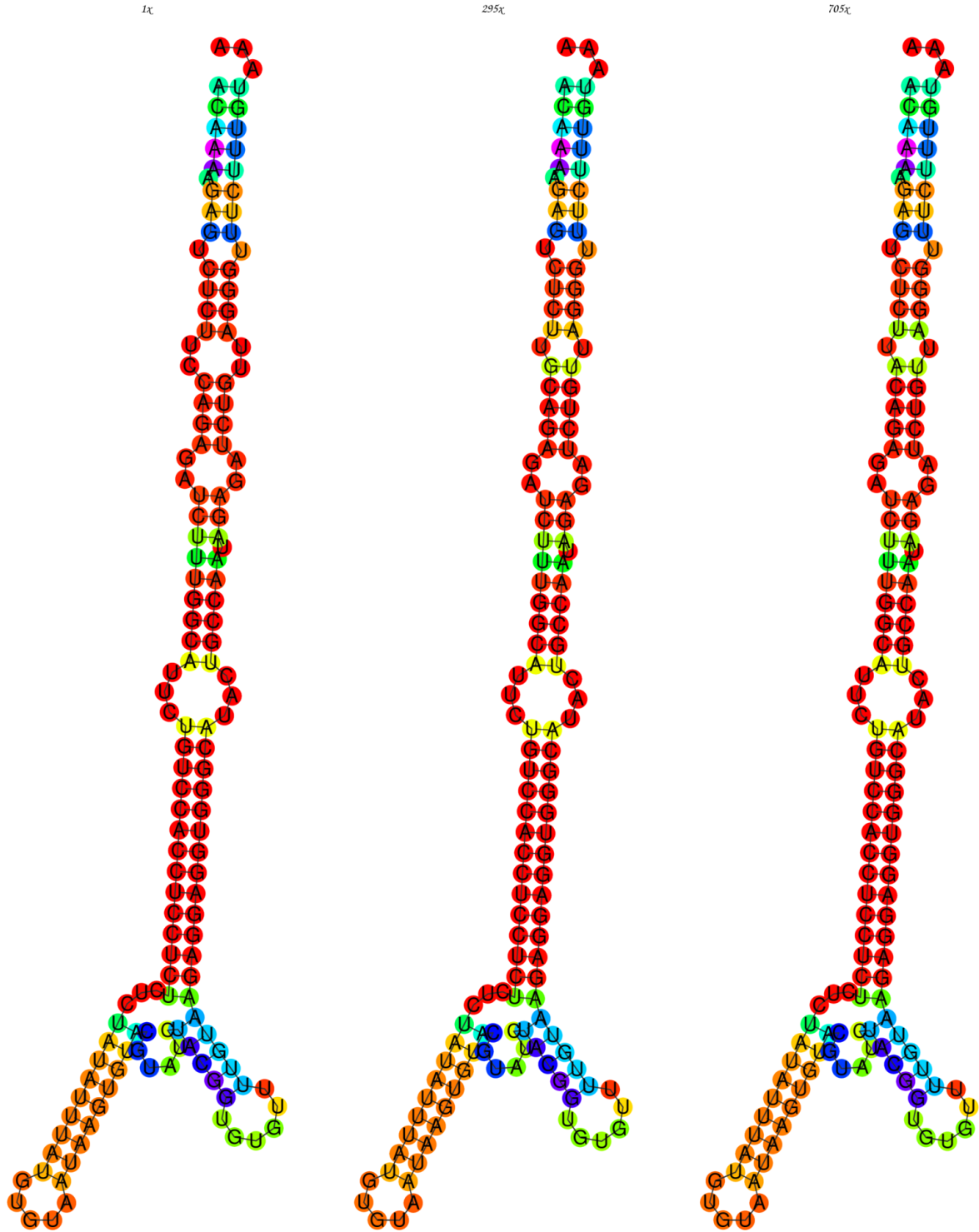


Figure 20: Secondary structures of three haplotypes of the bidirectionally-processed pri-miR394b. The base of the pri-miRNA is located at the top and the loop at the bottom. Terminal branched loops can be observed in all three structures. The miRNA-5p is consistently located on the left strand, directly under the third internal loop from the top, with the sequence 5'-UUGGCAUUCUGUCCACCUC-3'. The only SNP occurs at position 16 from the 5' end, resulting in a larger internal loop, located 9 nt from the 5' end of the miRNA-5p towards the base, in the left haplotype.

4 Discussion

Within this section, the reasons and implications of the observed results are discussed in detail. Furthermore, the challenges and limitations of the analyses conducted in this thesis are examined. To maintain clarity, the subsections follow the same order as in the previous sections.

Genotype heatmaps

The genotype heatmaps show the variants at each pri-miRNA position across the 1135 accessions of *A. thaliana* for every pri-miRNA. Consequently, variants located within the miRNA-5p and miRNA-3p regions are also displayed. However, annotating 5p and 3p regions alone does not reveal which strand participates in post-transcriptional gene regulation, as the miRNA* strand is normally degraded [40]. Therefore, incorporating information on strand degradation would be useful to enhance the interpretation of variant occurrences within the miRNA-5p and miRNA-3p. It should be noted that even the miRNA* strand can be loaded onto AGO proteins [40].

As variants are displayed in red, red bars in the genotype heatmaps appear very promising at first glance. Nevertheless, these variants at a specific position could all be identical which would indicate a highly conserved position. Therefore, this has to be taken into account when interpreting the genotype heatmaps. Of course, these variants could differ from one another showing variable positions but this has to be verified by looking at the VCF file.

Plots per processing type

For the creation of the aggregated heatmaps and the variant frequency plots, the loop and miRNA regions of the individual pri-miRNAs were scaled (see Subsection 2.8). While scaling the miRNA regions introduced minimal bias due to their highly conserved length that varies only a few nucleotides, scaling the loop regions presented significant challenges due to their high variability in length. Scaling long loops down using block aggregation with the maximum operator artificially increased the variant density. Conversely, scaling short loops up with nearest-neighbor resampling duplicates positions, resulting in single variants to occur in multiple adjacent positions and therefore creating artificial variant clusters. Moreover, for loops shorter than 41 nucleotides, it is not possible to scale the central region of the loop as described in Subsection 2.8 because there is no central region left when subtracting 40. In these cases the entire loop had to be scaled, introducing additional bias into the regions that contain important rulers for pri-miRNA processing.

In general, the comparison of the plots across processing types is limited by the varying number of pri-miRNAs in each group (see Table 2). For example, the higher number of variant positions observed in the bidirectional group compared to the SLTB group among the non-predicted miRNAs can be attributed to group size differences, as the bidirectional group contains 28 pri-miRNAs, while the SLTB group comprises only four.

As with the genotype heatmaps, a high variant frequency at a position in the variant frequency plots does not necessarily indicate high sequence variability but could also show a highly conserved region if these variants are all identical.

Combining the analyses for the non-predicted and the predicted miRNAs would likely not reveal any new patterns in the aggregated heatmaps because they would almost entirely be blue. For the variant frequency plots, merging non-predicted and predicted miRNAs would also result in a simple overlay of the plots, offering limited additional value.

Statistical analyses

The permutation test was performed to analyze whether variants are more abundant in miRNA regions compared to the whole genome. The limitation of the test performed within this thesis is that it does not account for the variant frequency at a position, meaning that a variant occurring in a single accession is weighted equally to one observed in 40% of the accessions. Filtering variant positions based on a MAF or missing rate would likely alter the results, leading to fewer pri-miRNAs showing a higher variant abundance than expected by chance.

To examine the varying number of variants across the regions of the pri-miRNAs for different processing types, a Kruskal-Wallis test was performed. Specifically, the high proportion of variants in the miRNA-5p region within the LTB group of the non-predicted miRNAs (see Table 5) appeared significantly different at first glance. Nevertheless, no statistical significance was observed, most likely because 12 out of the 17 pri-miRNAs belonging to the LTB group of the non-predicted miRNAs show almost no variants in the miRNA-5p region, indicating that the high average proportion is driven by only five pri-miRNAs. Furthermore, the miRNA regions of the predicted miRNAs appeared less conserved compared to those of the non-predicted miRNAs (see Table 5 and Table 6). This pattern could be caused by inaccurate predictions of the miRNA regions by StarSeeker. This hypothesis is supported by the fact that the variant frequency plots of the predicted miRNAs (see Figure 12) show high variant frequencies at the ends of the miRNA regions, indicating that shifting the predicted miRNA regions a few nucleotides could result in fewer variants within the miRNA regions. Combining the data of the non-predicted and the predicted miRNAs would increase the sample size and thereby enhance the statistical power of the Kruskal-Wallis test to detect significant differences. For this reason, this should be considered in future research.

Structure prediction of pri-miRNAs

The prediction of the pri-miRNA structures is based on the transcript derived from the region defined in Figure 4. Therefore, the base regions are each 25 nt long. *In vivo*, the lengths of these base regions may vary, leading to different secondary structures. Moreover, the RNA secondary structure is influenced by additional factors such as proteins or RNA modifications (see Subsection 1.2), which could also result in *in vivo* structures that differ from the predicted ones. Consequently, the predicted structures may not reflect the actual *in vivo* structure.

In the predicted structures, some nucleotides overlap (e.g., in Figure 16). This occurs because these structures are displayed in two dimensions, whereas in reality, they fold into three dimensional structures. Moreover, open loops are visible (see Figure 17) because these positions represent the 5' and 3' ends of the RNA strand, which are not connected by RNAfold [10, 18].

As described in Subsection 1.2, the 9–11 nt ruler as well as the 15–17 nt ruler are crucial for pri-miRNA processing. In every structure from Figure 16 to Figure 20, at least one internal loop falls within the range of the 9–11 nt ruler or the 15–17 nt ruler. It is highly likely, that these internal loops actually represent these rulers. Furthermore, in the miRNA/miRNA* duplex the presence of GC-rich motifs at position 8–9 and 18–19 is important for precise pri-miRNA processing (see Subsection 1.2). In the structure of pri-miR156a (Figure 16) these motifs occur at positions 9 and 18, in pri-miR173 (Figure 17) and pri-miR841a (Figure 18) they occur at positions 8 and 19, in pri-miR391 (Figure 19) at positions 8 and 18, and in pri-miR394b (Figure 20) at position 19.

The following text formulates hypotheses on how processing of the pri-miRNAs shown in Figure 17 to Figure 20 is affected by the variants. Figure 17 shows the secondary structures of the BTL-processed pri-miR173. The variant located in the stem of this pri-miRNA can affect the second cut precision, while the variant in the loop region may shift the processing mechanism to LTB [28]. Figure 18 and Figure 19 present the secondary structures of the LTB-processed pri-miR841a and the bidirectionally-processed pri-miR391. The rulers of both pri-miRNAs are affected by variants, making an impact on processing likely. Specifically, the size of an internal loop within the loop region is altered in pri-miR841a, whereas in pri-miR391 the internal loop size within the base region is changed. Figure 20 shows three haplotype structures of the bidirectionally-processed pri-miR394b, two of which exhibit the same structure despite a SNP in the base region. In contrast, the third haplotype shows an altered internal loop located within the base region, which may influence processing. Structural variants do not necessarily change the processing type of a pri-miRNA but can also lead to inaccurate processing, causing the production of alternative miRNAs (isomiRs) (see Subsection 1.2). To examine the effects of these variants on processing in detail, further research is necessary.

5 Conclusion

This thesis examined the presence and impact of natural sequence variation in pri-miRNAs across the 1135 *A. thaliana* accessions. The created genotype heatmaps visualize the presence and distribution of sequence variants per pri-miRNA and show great variety among pri-miRNAs.

The aggregated heatmaps and the variant frequency plots were generated to identify potential patterns between the processing types. Although the aggregated heatmaps did not reveal distinct patterns between the processing types, they show that variants are not positionally consistent across multiple pri-miRNAs within a processing type. Moreover, the variant frequency plots indicate that the mature miRNA regions are less affected by sequence variation than base and loop regions, particularly among non-predicted miRNAs. Statistical analyses revealed 11 pri-miRNAs with a significantly lower abundance of variants than expected by chance and 29 pri-miRNAs with a significantly higher abundance. However, no statistically significant differences were observed across the processing types.

The haplotype sequences for each accession and pri-miRNA were reconstructed and their variants were visualized using haplotype heatmaps. These haplotype heatmaps show that the presence and distribution of variants differ widely among haplotypes and pri-miRNAs. Moreover, the number of haplotype sequences varies greatly among pri-miRNAs. Finally, the secondary structures for all haplotypes were predicted to assess the structural consequences of the variants. Key structural features relevant for pri-miRNA processing as well as sequence motifs were observed, allowing for hypotheses on how specific sequence variants may affect pri-miRNA processing.

Future work

In August 2026, version 23 of miRBase was released. This update introduces new miRNAs, removes invalid entries and updates existing names and sequences. Consequently, all analyses performed within this thesis could be repeated using this updated dataset to include additional miRNAs and to improve data precision.

To verify the hypotheses of this work on how specific sequence variants may affect pri-miRNA processing, *in vitro* dicing assays [35] could be performed for the candidate pri-miRNAs.

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Declaration of authorship

I hereby confirm that I have written this thesis independently, that I have used no other sources or aids than those indicated, and that I have observed the general principles of good scientific practice. Furthermore, I declare that this work has not been published, nor has it been submitted elsewhere for an academic degree.

Moreover, I confirm that I have only used generative artificial intelligence as an aid, and that the work submitted is predominantly the result of my input. I am aware that the use of artificial intelligence does not guarantee the quality of the content. I take full responsibility for the adoption of any artificial intelligence generated content. For this thesis, Gemini and ChatGPT were used to a limited extent to generate source code, to paraphrase individual sentences, and to check style, spelling, and grammar.

Halle (Saale), 14th September 2026

Place, Date



Signature